US 20250279519A1

(19) United States
(12) Patent Application Publication
SHI et al.
(10) Pub. No.: US 2025/0279519 A1
(43) Pub. Date: Sep. 4, 2025

(54) BATTERY AND ELECTRICAL DEVICE

(71) Applicant: **CONTEMPORARY AMPEREX TECHNOLOGY (HONG KONG) LIMITED**, Hong Kong (CN)

(72) Inventors: **Weigang SHI**, Ningde (CN); **Lifei ZHANG**, Ningde (CN); **Xiaofang WANG**, Ningde (CN); **Zhi WANG**, Ningde (CN); **Wenchao LUO**, Ningde (CN); **Si SHI**, Ningde (CN); **Hao SUN**, Ningde (CN); **Runquan CHENG**, Ningde (CN); **Yanlin ZHU**, Ningde (CN)

(73) Assignee: **CONTEMPORARY AMPEREX TECHNOLOGY (HONG KONG) LIMITED**, Hong Kong (CN)

(21) Appl. No.: **19/209,864**

(22) Filed: **May 16, 2025**

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2023/116268, filed on Aug. 31, 2023.

(30) **Foreign Application Priority Data**

Dec. 30, 2022 (WO) PCT/CN2022/144191

Publication Classification

(51) **Int. Cl.**

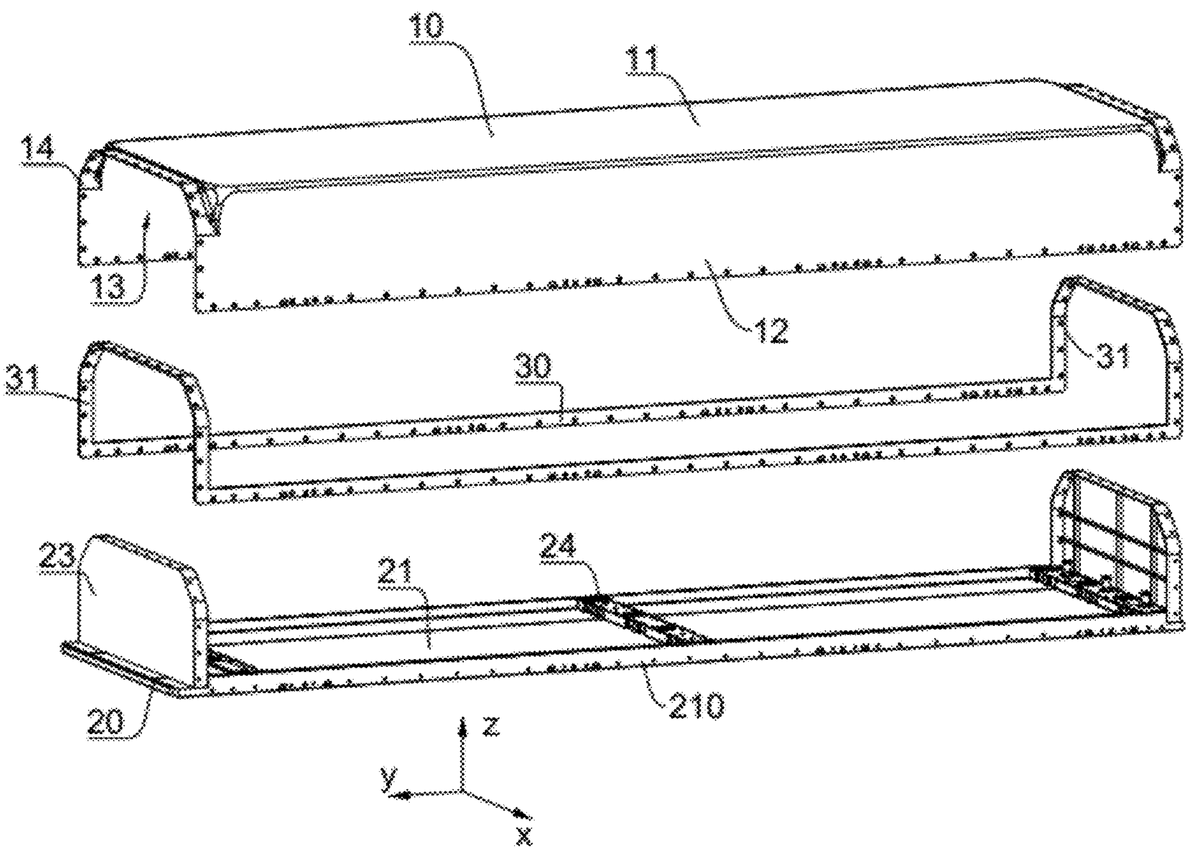
H01M 50/233	(2021.01)
H01M 50/224	(2021.01)
H01M 50/244	(2021.01)
H01M 50/262	(2021.01)
H01M 50/278	(2021.01)

(52) **U.S. Cl.**

CPC ***H01M 50/233*** (2021.01); ***H01M 50/224*** (2021.01); ***H01M 50/244*** (2021.01); ***H01M 50/262*** (2021.01); ***H01M 50/278*** (2021.01)

(57) **ABSTRACT**

A battery and an electrical device. The battery includes: a battery cell; a first box, where the first box includes a first end wall and a first side wall; and a second box, where the second box and the first box are connected to each other to jointly enclose and form a closed space for accommodating the battery cell, the second box includes a second end wall, the second end wall is disposed opposite to the first end wall along a first direction, the second box has a first side surface in a second direction, and the first direction intersects with the second direction. One end of the first side wall is connected to the first end wall, and the other end of the first side wall is connected to the first side surface.



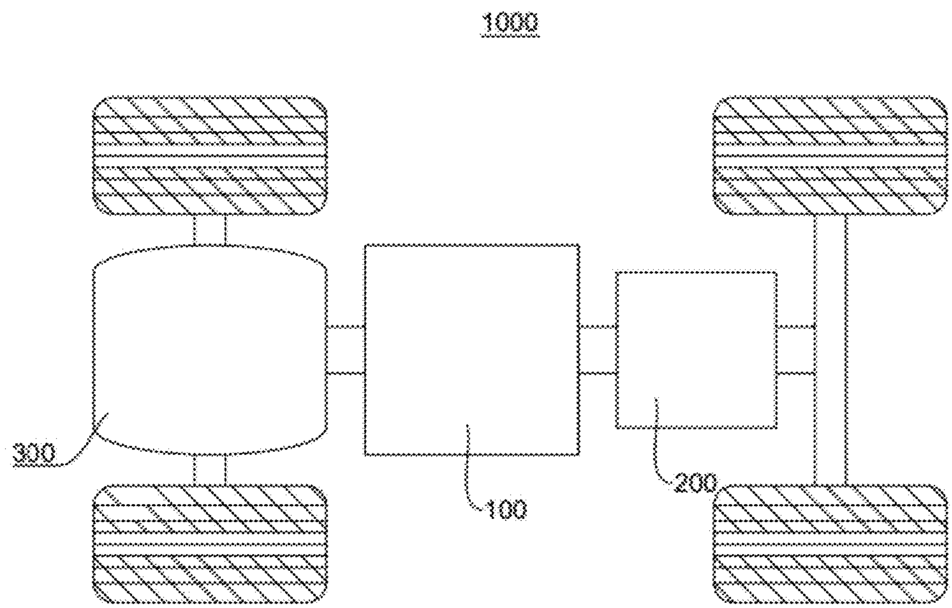


FIG. 1

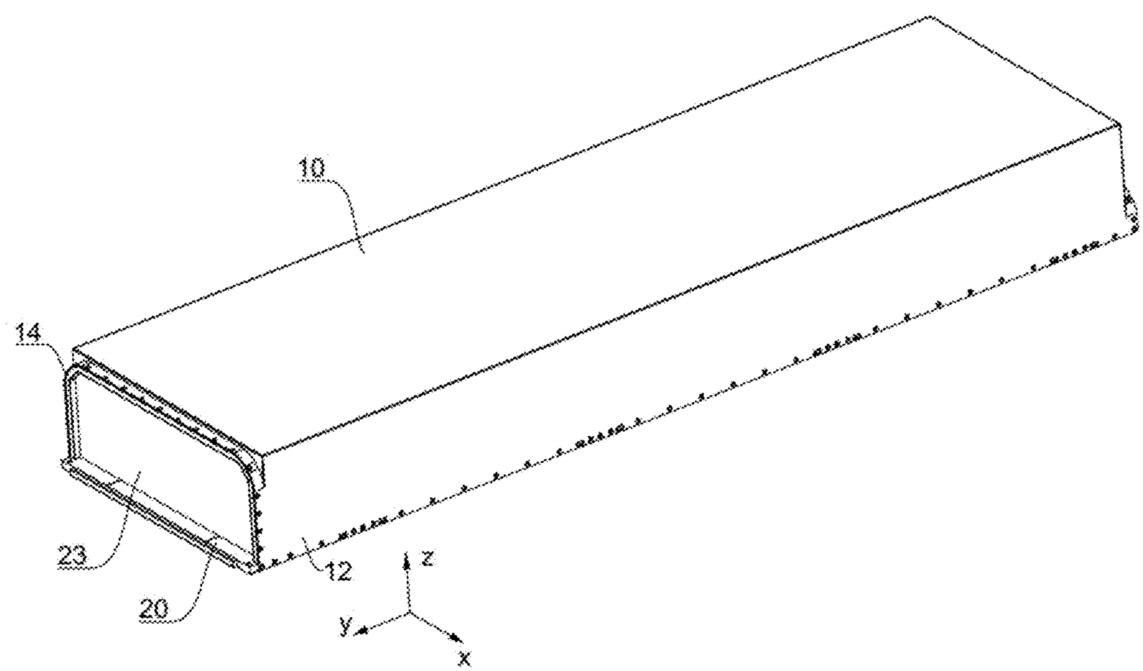


FIG. 2

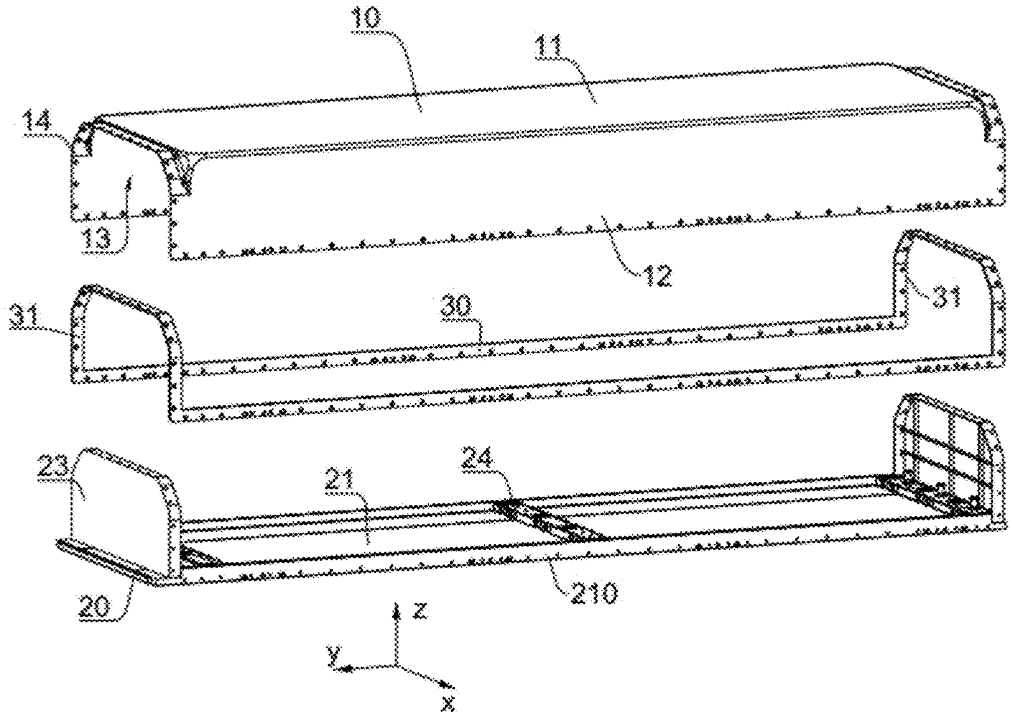


FIG. 3

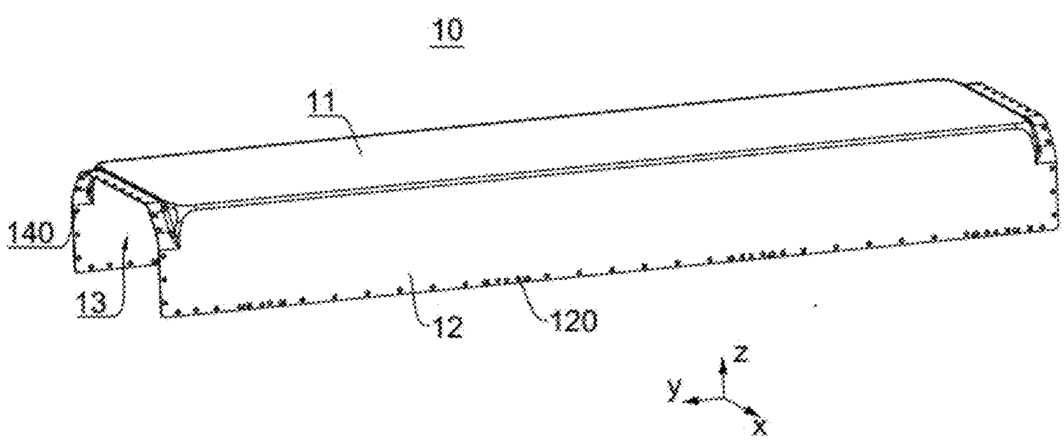


FIG. 4

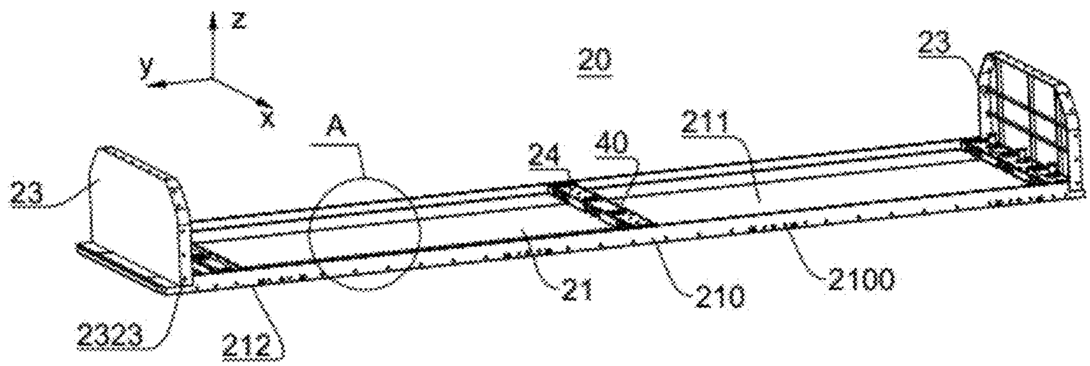


FIG. 5

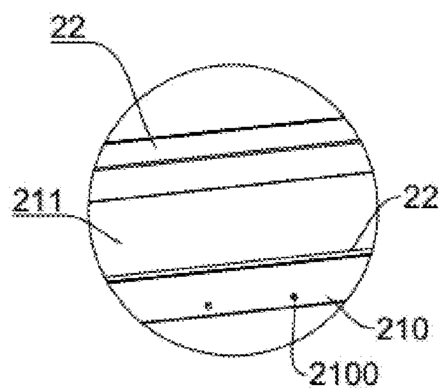


FIG. 6

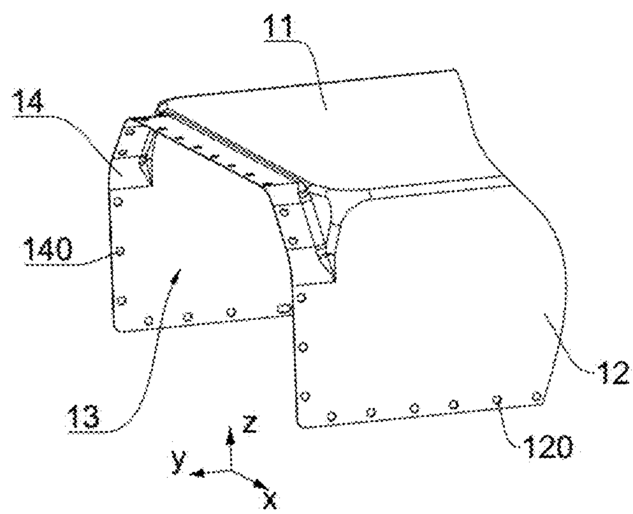


FIG. 7

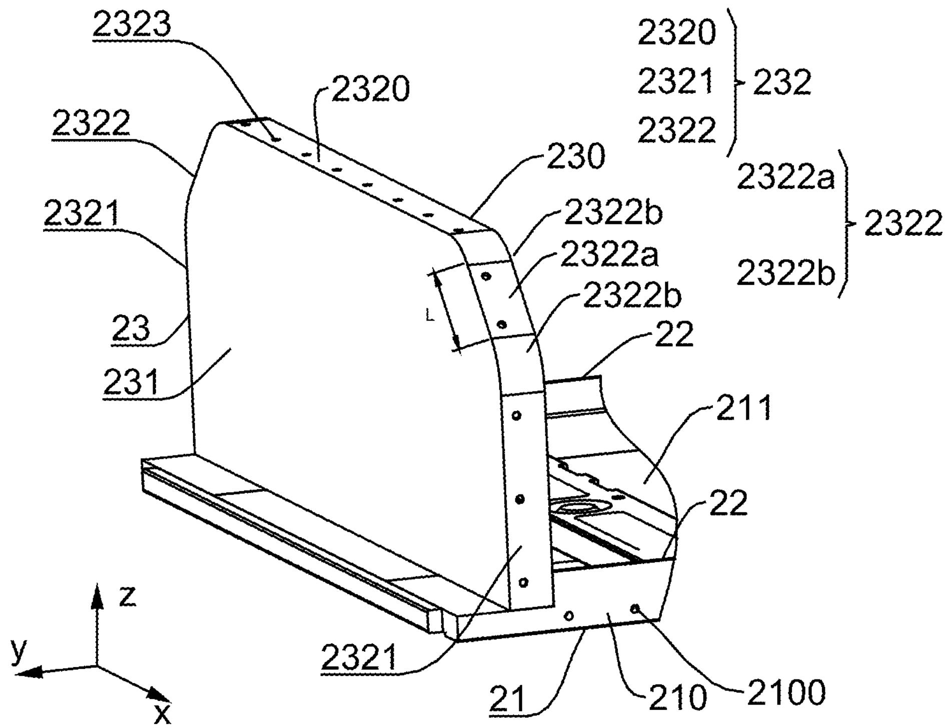

2323
2320
2322
2321
23
231
230
2322b
2322a
2322b
L
22
211
22
2321
21
210
2100
z
y
x
2320
2321
2322
232
2322a
2322b
2322

FIG. 8

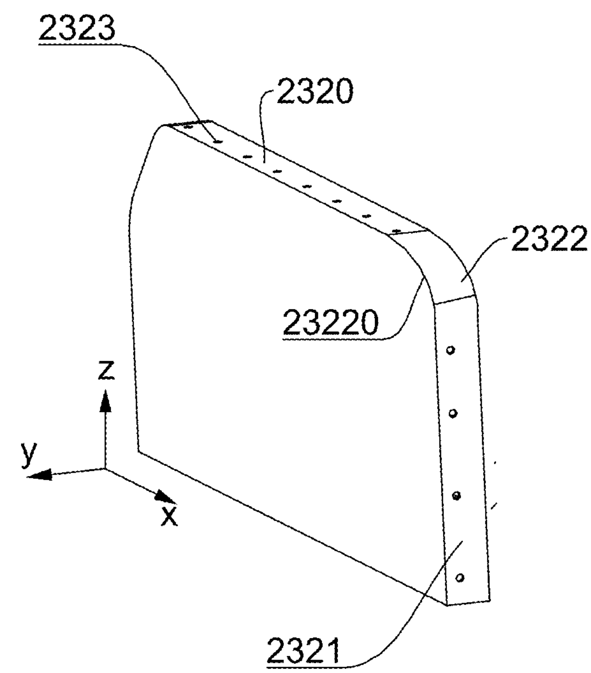

2323
2320
2322
23220
z
y
x
2321

FIG. 9

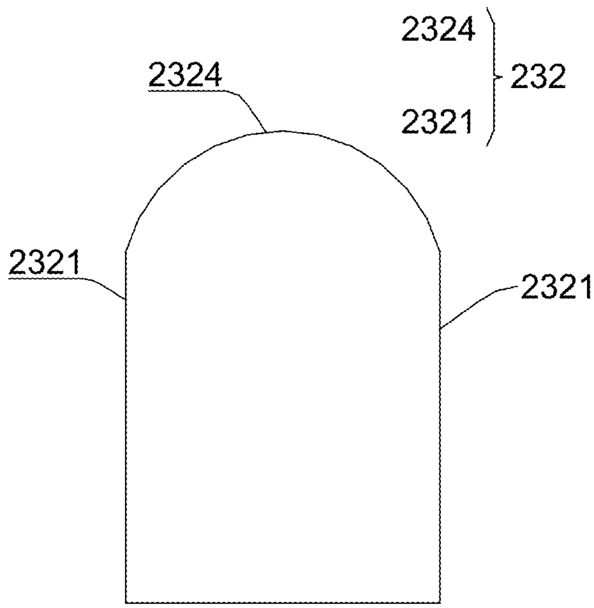


FIG. 10

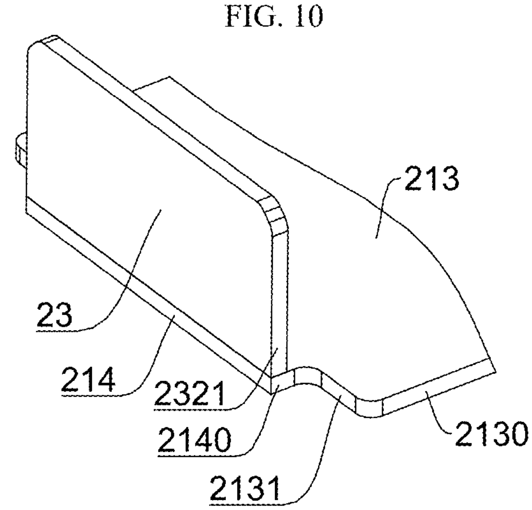


FIG. 11

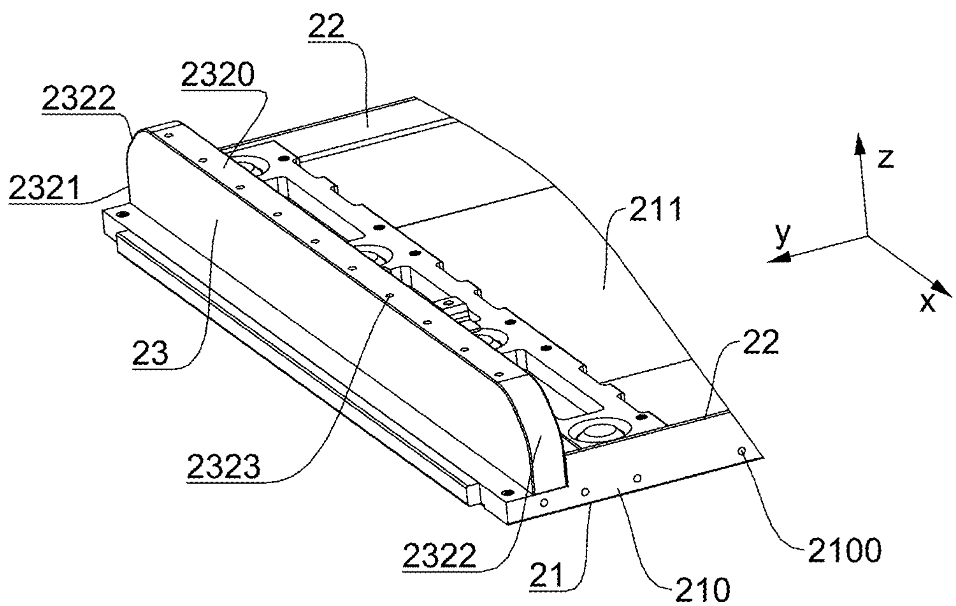

22
2320
2322
2321
211
z
y
x
23
22
2323
2100
2322
21
210

FIG. 12

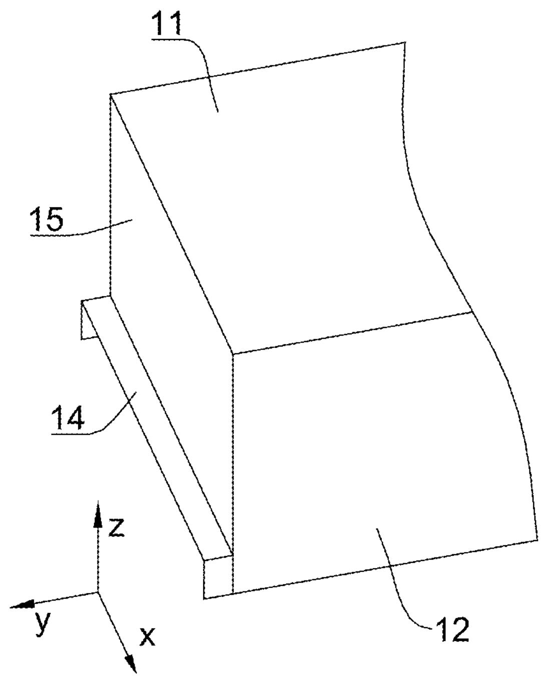

11
15
14
z
y
x
12

FIG. 13

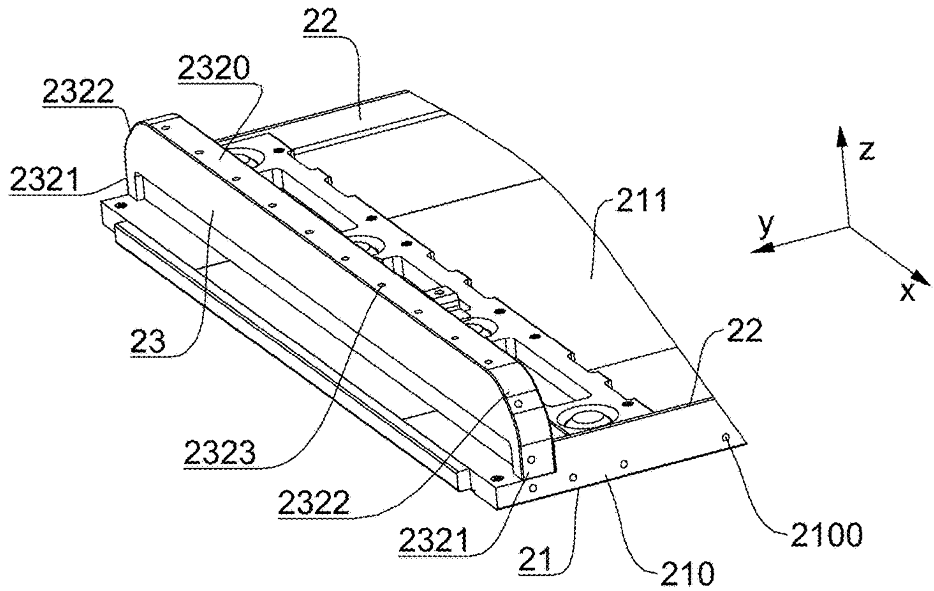

22
2320
2322
2321
211
z
y
x
23
22
2323
2322
2321
21
210
2100

FIG. 14

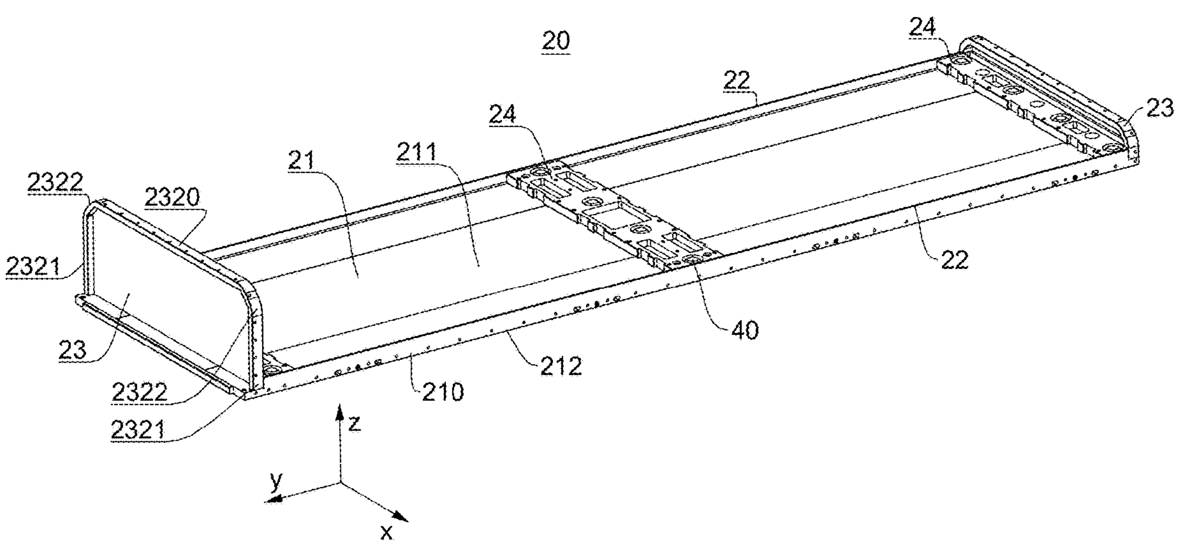

20
24
22
23
24
21
211
2322
2320
2321
23
2322
2321
210
212
40
22
z
y
x

FIG. 15

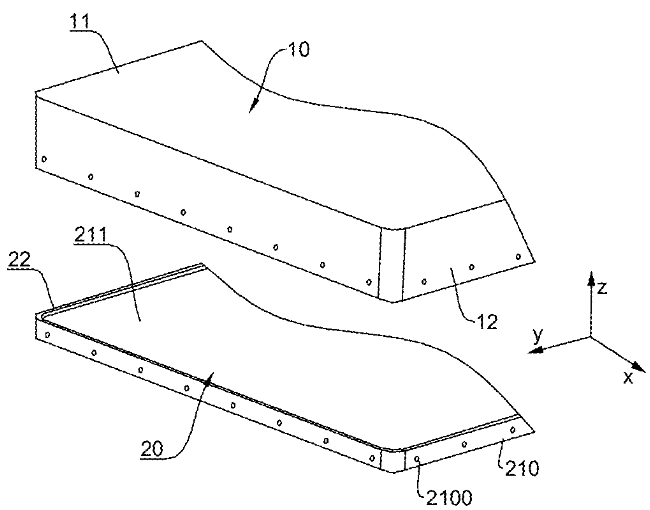


FIG. 16

BATTERY AND ELECTRICAL DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of PCT Patent Application PCT/CN2023/116268 filed on Aug. 31, 2023, which claims priority to PCT Patent Application PCT/CN2022/144191 entitled "BATTERY AND ELECTRICAL DEVICE" filed on Dec. 30, 2022, the entire contents of each of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] This application relates to the technical field of batteries, and in particular, to a battery and an electrical device.

BACKGROUND

[0003] Energy saving and emission reduction are the key to the sustainable development of the automotive industry, and electric vehicles have become an important part of the sustainable development of the automotive industry by virtue of their advantages of energy saving and environmental protection. For electric vehicle, battery technology is a crucial factor driving the development thereof.

[0004] In the development of battery technology, how to increase the energy density of batteries is an urgent technical problem that needs to be solved in battery technology.

SUMMARY

[0005] This application provides a battery and an electrical device that can increase the energy density of the battery.

[0006] This application is achieved by means of the following technical solution:

[0007] In a first aspect, this application provides a battery including: a battery cell; a first box, where the first box includes a first end wall and a first side wall; and a second box, where the second box and the first box are connected to each other to jointly enclose and form a closed space for accommodating the battery cell, the second box includes a second end wall, the second end wall is disposed opposite to the first end wall along a first direction, the second box has a first side surface in a second direction, and the first direction intersects with the second direction, where one end of the first side wall is connected to the first end wall, and the other end of the first side wall is connected to the first side surface.

[0008] In the above solutions, by connecting one end of the first side wall of the first box to the first side surface of the second box to achieve connection between the first box and the second box, the connection between the first box and the second box can be achieved without disposing a flange structure protruding along the second direction, thereby improving the space utilization rate of the battery in the second direction to accommodate more battery cells or reducing the volume of the battery, and thus improving the volumetric energy density of the battery.

[0009] According to some embodiments of this application, an inner surface of the first side wall is connected to the first side surface.

[0010] In the above solutions, by connecting the inner surface of the first side wall (i.e. a surface that is of the first side wall and that is facing the first side surface) to the first side surface, a size of the battery in the second direction can be effectively reduced to increase the volumetric energy density of the battery as much as possible.

[0011] According to some embodiments of this application, the second end wall has a first surface facing the first end wall, and one end that is of the first side wall and that is away from the first end wall exceeds the first surface along a direction of the first end wall pointing towards the second end wall.

[0012] In the above solutions, the end that is of the first side wall and that is away from the first end plate exceeds the first surface, so that the first side wall may be directly connected to the first side surface, and on the one hand, the assembly difficulty and the manufacturing cost caused by the indirect connection of the first side wall to the first side surface through an intermediate connector can be effectively reduced, and on the other hand, an inner surface of a portion of the first end wall exceeding the first surface may be connected to the first side surface, and a size of the battery in the second direction can be effectively reduced to increase the volumetric energy density of the battery as much as possible.

[0013] According to some embodiments of this application, the second end wall has a second surface facing away from the first end wall, and the end that is of the first side wall and that is away from the first end wall does not exceed the second surface along a direction of the first end wall pointing towards the second end wall.

[0014] In the above solutions, the end that is of the first side wall and that is away from the first end wall does not exceed the second surface, so that a size of the battery in the first direction is controlled, and enabling the battery to have a higher volumetric energy density.

[0015] According to some embodiments of this application, a material density of the first box is lower than a material density of the second box.

[0016] In the above solutions, since the first side wall is connected to the first side surface, the proportion of the first box to the battery is increased, and the material density of the first box is smaller than that of the second box, the density of the battery is reduced, and the weight of the battery is reduced under the same volume, thus increasing the weight energy density of the battery.

[0017] According to some embodiments of this application, the material of the first box is plastic, and the material of the second box is aluminum alloy.

[0018] In the above solutions, the first box may be an upper box of the battery, the second box may be a lower box of the battery, and the second box is made of an aluminum alloy material, so that the second box has higher structural strength and rigidity, and plays a greater protective role for the battery cell. As an upper box, the first box plays a role of closing the battery cell and the upper and lower boxes, and it is made of plastic, so that the manufacturing cost of the battery can be effectively reduced, and the weight of the battery can be effectively reduced to improve the weight energy density of the battery.

[0019] According to some embodiments of this application, the battery further includes a first fastener, and the first side wall is connected to the first side surface through the first fastener.

[0020] In the above solutions, the first fastener is disposed to connect the first side wall and the first side surface, so that there is higher connection stability between the first side wall and the first side surface, and the structural stability of the battery is improved.

[0021] According to some embodiments of this application, the first side wall is provided with a first through hole, the first side surface is provided with a first threaded hole, and the first fastener passes through the first through hole and is connected to the first threaded hole.

[0022] In the above solutions, the first fastener may be a connecting component with an external thread such as a bolt and a screw, which can be effectively connected to the first threaded hole, improve the connection stability between the first side wall and the first side surface, and allow good hermeticity between the first side wall and the first side surface.

[0023] According to some embodiments of this application, the battery further includes a first sealing element, and the first sealing element is disposed between the first side wall and the first side surface.

[0024] In the above solutions, by disposing a first sealing element between the first side wall and the first side surface, the hermeticity between the first side wall and the first side surface can be improved, thus improving the hermeticity of the battery.

[0025] According to some embodiments of this application, a quantity of the first side walls is two, and the two first side walls are disposed opposite to each other along the second direction; the second end wall is located between the two first side walls, and the second end wall has the two first side surfaces disposed opposite to each other along the second direction; and the first side surface corresponds one-to-one to the first side wall.

[0026] In the above solutions, the quantity of the first side walls is two, and the two first side walls are disposed opposite to each other along the second direction and are respectively connected to corresponding first side surfaces, which can effectively improve the space utilization rate of the battery in the second direction, thus increasing the volumetric energy density of the battery.

[0027] According to some embodiments of this application, the second box further includes two second side walls, the two second side walls are disposed opposite to each other along the second direction and connected to the second end wall, and the battery cell is disposed between the two second side walls.

[0028] In the above solutions, two opposite second side walls are disposed on the second end wall along the second direction. On the one hand, the structural strength of the second box can be improved, on the other hand, a mounting region of the battery cell can be defined, so that the battery cell can be stably assembled in the second box, and on the other hand, since the second side wall protrudes from the surface of the second end wall, when the battery cell is bonded to the second box by glue, a glue overflow space can be effectively controlled, and the risk of waste and environmental pollution caused by glue overflow can be reduced.

[0029] According to some embodiments of this application, openings are respectively formed at two ends of the first box along a third direction; the second box further includes two third side walls, the two third side walls are disposed opposite to each other along the third direction and connected to the second end wall, and the two third side walls respectively close the two openings; and the first direction, the second direction and the third direction intersect in pairs.

[0030] In the above solutions, by disposing a third side wall, on the one hand, components such as explosion-proof valves, water-cooled connectors, or high- and low-voltage plug-in connectors can be mounted on the third side wall to enable the battery to perform normal charging and discharging operations; and on the other hand, compared with a flange structure protruding along the third direction disposed between the first box and the second box, by disposing the third side wall to close an opening, the space utilization rate of the battery in the third direction can be improved, and the volumetric energy density of the battery can be increased.

[0031] According to some embodiments of this application, the third side wall includes a third surface facing the closed space, a fourth surface facing away from the closed space and a second side surface connecting the third surface and the fourth surface; and the first box includes two connecting portions, the two connecting portions are respectively located at the two ends of the first box along the third direction, the connecting portion peripherally forms the opening, and the connecting portion is connected to the second side surface.

[0032] In the above solutions, by disposing a connecting portion to be connected with the second side surface of the third side wall, the connection stability and hermeticity of the first box and the second box can be improved.

[0033] According to some embodiments of this application, the second side surface includes a first flat surface, a second flat surface and a transition surface, the first flat surface is disposed at one end that is of the third side wall and that is facing away from the second end wall, the two second flat surfaces are respectively disposed at two ends of the third side wall along the second direction, and the transition surface connects the first flat surface and the second flat surface, so that the first flat surface smoothly transitions to the second flat surface.

[0034] In the above solutions, the first flat surface and the second flat surface are non-coplanar surfaces. Therefore, by disposing the transition surface, a smooth transition between the first flat surface and the second flat surface is allowed, which can be conducive to forming a good sealing surface between the connecting portion and the second side surface, enabling the battery to have higher hermeticity.

[0035] According to some embodiments of this application, the transition surface includes a circular arc transition surface.

[0036] In the above solutions, by disposing the transition surface as the circular arc transition surface, the first flat surface can smoothly transition to the second flat surface, so that a sealing structure can be smoothly disposed between the third side wall and the first box, reducing the risk of sealing failure due to damage to the sealing structure caused by stress concentration of the sealing structure, which is conducive to the sealing performance between the second side surface and the connecting portion, and improving the hermeticity and reliability of the battery.

[0037] According to some embodiments of this application, the transition surface has a first edge in the third direction, the first edge is circular arc-shaped, and a radius of the first edge is greater than or equal to 10 mm.

[0038] The first edge may correspond to an extension trajectory of the transition surface, and the first edge may also refer to an edge of the transition surface parallel to its extension direction. In the above solutions, by defining the first edge of the transition surface in a circular arc shape, i.e., defining the first edge as a circular arc line, the transition surface extends in an arc-shaped trajectory, and by disposing the radius of the first edge to be greater than or equal to 10 mm, the first flat surface can smoothly transition to the second flat surface, and the sealing structure can be gently disposed between the third side wall and the first box, effectively reducing the risk of sealing failure due to damage to the sealing structure caused by stress concentration of the sealing structure, which is conducive to the sealing performance between the second side surface and the connecting portion, and improving the hermeticity and reliability of the battery.

[0039] According to some embodiments of this application, the transition surface includes a first transition surface and a second transition surface connected to each other, an edge of the first transition surface in the third direction is flat, and an edge of the second transition surface in the third direction is circular arc-shaped.

[0040] In the above solutions, by disposing the first transition surface and the second transition surface, on the one hand, the first flat surface can smoothly transition to the second flat surface, which is conducive to improving the sealing performance between the second side surface and the connecting portion, and improving the hermeticity and reliability of the battery; and on the other hand, the first transition surface can provide a flat surface for locking of the third side wall and the first box, so that a locking piece (such as a bolt) can pass through the first box and be locked into the first transition surface stably and with good hermeticity, thereby improving the structural stability of the battery, and enabling the battery to have higher reliability.

[0041] According to some embodiments of this application, a quantity of the first transition surfaces is n, with $n \geq 1$, and a quantity of the second transition surfaces is n+1; and any one of the first transition surfaces is disposed between two adjacent second transition surfaces.

[0042] In the above solutions, by disposing the quantity of second transition surfaces to be one more than that of the first transition surfaces and disposing any one of the first transition surfaces between the two adjacent second transition surfaces, on the one hand, the first flat surface can smoothly transition to the second flat surface, that is, the transition surfaces connected to the first flat surface and the second flat surface are both the second transition surfaces with arc surfaces, reducing the risk of sealing failure caused by formation of edges and corners between the flat surfaces. On the other hand, by disposing the first transition surface, a plurality of locking positions can be provided between the third side wall and the first box, which is conducive to effective connection of the locking piece to the third side wall and the first box, improving the structural stability of the battery, and enabling the battery to have higher reliability.

[0043] According to some embodiments of this application, a length of the first transition surface is greater than or equal to 6 mm.

[0044] In the above solutions, by disposing the length of the first transition surface to be greater than or equal to 6 mm, one or more locking positions can be disposed on the first transition surface, which is conducive to effective connection of the locking piece to the third side wall and the first box, improving the structural stability of the battery, and enabling the battery to have higher reliability.

[0045] According to some embodiments of this application, the second side surface includes a circular arc surface and a second flat surface, the two second flat surfaces are respectively disposed at two ends of the third side wall along the second direction, and one end of one of the two second flat surfaces and one end of the other of the two second flat surfaces face away from the second end wall and are connected through the circular arc surface.

[0046] In the above solutions, a smooth transition between two second flat surfaces disposed opposite to each other through the circular arc surface can be conducive to forming a good sealing surface between the connecting portion and the second side surface, enabling the battery to have higher hermeticity.

[0047] According to some embodiments of this application, the second flat surface is flush with the first side surface.

[0048] In the above solutions, the second flat surface is flush with the first side surface, which can reduce the risk of generating a gap between the first side wall and the second side surface, reduce the risk of generating a gap between the connecting portion and the second flat surface, and thus improve the hermeticity of the battery.

[0049] According to some embodiments of this application, the second end wall includes a first portion and a second portion, and the second portion is located on one side of the first portion and connected to the first portion along the second direction; and a size of the first portion in the second direction is greater than a size of the second portion in the second direction.

[0050] In the above solutions, the first end wall includes a first portion and a second portion. By disposing the size of the first portion in the second direction to be greater than the size of the second portion in the second direction, a notch is formed between the first portion and the second portion, or in other words, an end portion of the second end wall is narrowed. Since the end portion of the second end wall is not utilized by the battery cell, by narrowing the portion, on the one hand, the volume of the battery can be reduced, and the mounting space of the battery can be saved, and on the other hand, an effect of reducing the weight of the battery can be achieved, so that the battery is lightweight.

[0051] According to some embodiments of this application, the second portion has a third side surface in the second direction, and the third side surface is flush with the second flat surface.

[0052] In the above solutions, disposing the second flat surface to be flush with the third side surface of the second portion is conducive to hermeticity between the third side wall and the second end wall and the first box, reducing the risk of sealing failure between the first box and the second box due to a gap generated by the third side wall and the second end wall at the second portion, and enabling the battery to have higher reliability.

[0053] According to some embodiments of this application, the first portion has a fourth side surface in the second direction, and the third side surface protrudes from the fourth side surface along the third direction.

[0054] In the above solutions, the third side surface protrudes from the fourth side surface along the third direction, that is, the second flat surface that is flush with the third side surface also protrudes from the fourth side surface, enabling a notch to be formed at a portion of the first box and the second box corresponding to a portion between the third side surface and the fourth side surface, or it is understood that the first box and the second box are narrowed at the portion, which on the one hand, can reduce the volume of the battery and save the mounting space of the battery, and on the other hand, can achieve the effect of reducing the weight of the battery, so that the battery is lightweight.

[0055] According to some embodiments of this application, the first portion has a third flat surface in the third direction, and one end of the third flat surface is in circular arc transition connection with the third side surface, and the other end of the third flat surface is in circular arc transition connection with the fourth side surface along the second direction.

[0056] In the above solutions, the third flat surface is disposed between the third side surface and the fourth side surface. By disposing the one end of the third flat surface to be in circular arc transition connection with the third side surface and the other end to be in circular arc transition connection with the fourth side surface, the risk of damage to a sealing structure between the second end wall, the third side wall and the first box due to stress concentration can be reduced, which is conducive to the hermeticity between the first box and the second box, and enabling the battery to have higher reliability.

[0057] According to some embodiments of this application, the second side surface includes a first flat surface and a transition surface, the first flat surface is disposed at one end that is of the third side wall that is facing away from the second end wall, and an end portion of the first flat surface smoothly transitions to a side surface of the second end wall through the transition surface.

[0058] In the above solutions, the second side surface includes the first flat surface and the transition surface. The first flat surface can form a good seal with the connecting portion. By disposing the transition surface between the first flat surface and a side surface of the second end wall, a smooth transition between the first flat surface and the side surface of the second end wall can be achieved, which can be conducive to forming a good sealing surface between the first box and the second box, enabling the battery to have higher hermeticity.

[0059] According to some embodiments of this application, the first box further includes a fourth side wall, and the fourth side wall is adjacent to the first side wall; and one end of the fourth side wall is connected to the first end wall, the connecting portion is disposed at the other end of the fourth side wall, and the connecting portion protrudes from the fourth side wall in a direction facing away from the closed space.

[0060] In the above solutions, the first box may be an upper box, and the second box may be a lower box. In a manufacturing process of the battery, the material cost and density of the upper box may be lower than those of the lower box. Therefore, by disposing a fourth side wall to increase the proportion of the first box in the battery, the manufacturing cost of the battery is effectively reduced and the weight energy density of the battery is increased.

[0061] According to some embodiments of this application, the battery further includes a second fastener, and the connecting portion is connected to the second side surface through the second fastener.

[0062] In the above solutions, by disposing the second fastener to connect the second side surface and the connecting portion, the connection stability between the connecting portion and the second side surface is improved, and the structural stability of the battery is improved.

[0063] According to some embodiments of this application, the connecting portion is provided with a second through hole, the second side surface is provided with a second threaded hole, and the second fastener passes through the second through hole and is connected to the second threaded hole.

[0064] In the above solutions, the second fastener may be a connecting component with an external thread such as a bolt or a screw, which can be effectively connected to the second threaded hole, improve the connection stability between the connecting portion and the second side surface, and allow good hermeticity between the connecting portion and the second side surface.

[0065] According to some embodiments of this application, the battery further includes a second sealing element, and the second sealing element is disposed between the connecting portion and the second side surface.

[0066] In the above solutions, by disposing the second sealing element between the connecting portion and the second side surface, the hermeticity between the connecting portion and the second side surface can be improved, thus improving the hermeticity of the battery.

[0067] According to some embodiments of this application, the battery further includes a first sealing element, and the first sealing element is disposed between the first side wall and the first side surface, where two ends of the first sealing element are respectively connected to the two second sealing elements.

[0068] In the above solutions, the first sealing element and the second sealing element may be integrally formed or separately connected. By disposing the first sealing element and the second sealing element, the good hermeticity between the first box and the second box can be achieved, thereby improving the safety of the battery.

[0069] According to some embodiments of this application, a size of one of the third side walls along the first direction is smaller than a size of the other one of the third side walls along the first direction.

[0070] In the above solutions, the larger third side wall may be mounted with components such as explosion-proof valves, water-cooled connectors, or high- and low-voltage plug-in connectors to enable the battery to perform normal charging and discharging operations; and by reducing the size of the other one of the third side walls, the size of a portion of the first box corresponding to the third side wall can be adaptively increased, increasing the proportion of the first box to the battery (in this embodiment, the first box may be an upper box with a lower material cost and density), thereby lowering the manufacturing cost of the battery and increasing the weight energy density of the battery.

[0071] According to some embodiments of this application, the first box has a fifth side surface in the third direction, the second box has a sixth side surface in the third direction, the sixth side surface is disposed adjacent to the first side surface, the fifth side surface and the sixth side surface are connected to each other, and the first direction, the second direction and the third direction intersect in pairs.

[0072] In the above solutions, by disposing the sixth side surface to be connected to the fifth side surface, the connection between the first box and the second box can be achieved without disposing a flange structure protruding along the third direction, thereby improving the space utilization rate of the battery in the third direction to accommodate more battery cells or reducing the volume of the battery, and thus improving the volumetric energy density of the battery.

[0073] According to some embodiments of this application, the sixth side surface is in a circular arc transition to the first side surface.

[0074] In the above solutions, disposing the sixth side surface to be in a circular arc transition to the first side surface can be conducive to forming a good sealing surface between the second box and the first box, reducing the risk of damage to a sealing structure disposed between the second box and the first box due to stress concentration, and enabling the battery to have higher hermeticity and reliability.

[0075] According to some embodiments of this application, the second end wall is provided with a mounting portion for mounting the battery to an electrical device.

[0076] In the above solutions, the second end wall is provided with the mounting portion to achieve stable assembly of the battery, so that the battery can provide stable electrical energy.

[0077] In a second aspect, this application further provides an electrical device including the battery according to any one of the first aspect, where the battery is configured to provide electrical energy.

[0078] The above description is only an overview of the technical solutions of this application, in order to be able to more clearly understand the technical means of this application, the technical solutions may be implemented in accordance with the contents of the specification, and in order to make the above and other objectives, features and advantages of this application more apparent and understandable, the specific implementations of this application are thereby listed.

BRIEF DESCRIPTION OF DRAWINGS

[0079] To describe technical solutions in embodiments of this application more clearly, the following outlines the drawings to be used in the embodiments. Understandably, the following drawings show merely some embodiments of this application, and therefore, are not intended to limit the scope. A person of ordinary skill in the art may derive other related drawings from the drawings without making any creative efforts.

[0080] FIG. **1** is a schematic diagram of a vehicle in some embodiments of this application;

[0081] FIG. **2** is a schematic diagram of a battery in some embodiments of this application;

[0082] FIG. **3** is a stereoscopic exploded diagram of a battery in some embodiments of this application;

[0083] FIG. **4** is a schematic diagram of a first box in some embodiments of this application;

[0084] FIG. **5** is a schematic diagram of a second box in some embodiments of this application;

[0085] FIG. **6** is an enlarged view of A in FIG. **5**;

[0086] FIG. **7** is a partial schematic diagram of the first box in some embodiments of this application;

[0087] FIG. **8** is a schematic diagram of a third side wall in some embodiments of this application;

[0088] FIG. **9** is a schematic diagram of a first flat surface, a second flat surface and a transition surface in some embodiments of this application;

[0089] FIG. **10** is a schematic diagram of a third side wall in other embodiments of this application;

[0090] FIG. **11** is a partial schematic diagram of a second box in other embodiments of this application;

[0091] FIG. **12** is a schematic diagram of a partial structure of the second box in other embodiments of this application;

[0092] FIG. **13** is a partial schematic diagram of a first box in other embodiments of this application;

[0093] FIG. **14** is a partial schematic diagram of a second box in other embodiments of this application;

[0094] FIG. **15** is a schematic diagram of a second box in some other embodiments of this application; and

[0095] FIG. **16** is a schematic diagram of a first box and a second box in some embodiments of this application; and

[0096] reference numerals: **100**—battery; **1000**—vehicle; **200**—controller; **300**—motor; **10**—first box; **11**—first end wall; **12**—first side wall; **120**—first through hole; **13**—opening; **14**—connecting portion; **140**—second through hole; **15**—fourth side wall; **16**—fifth side surface; **20**—second box; **21**—second end wall; **210**—first side surface; **2100**—first threaded hole; **211**—first surface; **212**—second surface; **213**—first portion; **2130**—fourth side surface; **2131**—third flat surface; **214**—second portion; **2140**—third side surface; **22**—second side wall; **23**—third side wall; **230**—third surface; **231**—fourth surface; **232**—second side surface; **2320**—first flat surface; **2321**—second flat surface; **2322**—transition surface; **2322*a***—first transition surface; **2322*b***—second transition surface; **2324**—circular arc surface; **2323**—second threaded hole; **24**—crossbeam; **25**—sixth side surface; **30**—first sealing element; **31**—second sealing element; **40**—mounting portion; z—first direction; x—second direction; and y—third direction.

DETAILED DESCRIPTION OF EMBODIMENTS

[0097] Implementations of this application are described in further detail below in conjuncture with the accompanying drawings and embodiments. The detailed description of the following embodiments and the accompanying drawings are used for exemplifying the principles of this application, but are not to be used for limiting the scope of this application. That is, this application is not limited to the described embodiments.

[0098] Unless otherwise defined, all technical and scientific terms used herein bear the same meanings as what is normally understood by a person skilled in the technical field of this application. The terms used herein are merely intended to describe specific embodiments but not to limit this application. The terms "include" and "contain" and any variations thereof used in the specification, claims, and brief description of drawings of this application are intended as non-exclusive inclusion.

[0099] In the description of the embodiments of this application, the terms "first", "second", and the like are merely intended to distinguish between different objects, and shall not be understood as any indication or implication of relative importance or any implicit indication of the number, particular sequence or primary-secondary relationship of the technical features indicated. In the descriptions of the embodiments of this application, "a plurality of" means two or more, unless otherwise specifically defined.

[0100] An "embodiment" referred to herein implies that particular features, structures, or characteristics described in conjunction with an embodiment can be included in at least one embodiment of this application. The occurrence of this phrase in various positions in the specification does not necessarily refer to the same embodiment, nor is it an independent or alternative embodiment that is mutually exclusive from other embodiments. It is expressly and implicitly understood by the person skilled in the art that the embodiment described herein can be combined with other embodiments.

[0101] In the description of the embodiments of this application, the term "and/or" describes only an association relationship between associated objects and represents that three relationships may exist. For example, A and/or B may represent the following three cases: A exists, both A and B exist, and B exists. In addition, the character "/" in this specification generally indicates an "or" relationship between the associated objects.

[0102] In the description of the embodiments of this application, the term "a plurality of" refers to two or more (including two), similarly, "a plurality of groups" refers to two or more groups (including two groups), and "a plurality of pieces" refers to two or more pieces (including two pieces).

[0103] In the description of the embodiments of this application, a directional or a positional relationship indicated by the terms such as "length", "width", "thickness", "up", and "down" is a directional or positional relationship based on the illustration in the drawings, and is merely intended for ease or brevity of description of the embodiments of this application, but not intended to mean or imply that the indicated device or component is necessarily located in the specified direction/position or constructed or operated in the specified direction/position. Therefore, such terms are not to be understood as a limitation on the embodiments of this application.

[0104] In the descriptions of the embodiments of this application, unless otherwise specified and defined explicitly, the technical terms "mounting", "connection", "join", and "fastening" should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or an electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, an internal communication between two elements, or an interaction between two elements. A person of ordinary skill in the art can understand specific meanings of these terms in the embodiments of this application as appropriate to specific situations.

[0105] The battery mentioned in the embodiments of this application refers to a single physical module that includes one or more battery cells to provide a higher voltage and capacity. For example, the battery mentioned in this application may include one or more battery cells. The battery further includes a first box and a second box. The first box and the second box are connected to jointly enclose and form a closed space. The battery cells are disposed in the closed space to avoid liquid or other foreign matters affecting the charging or discharging of the battery cells.

[0106] In the development of battery technology, how to improve the energy density of batteries is an urgent technical problem that needs to be solved in battery technology. The applicant finds that at present, the edges of the first box and the second box of the battery are provided with flange structures protruding outward from the battery, and the first box and the second box are connected to each other through the flange structures to achieve hermetic connection. However, the protruding flange structures occupy additional space, resulting in a low space utilization rate of the battery, and affecting the volume energy density of the battery.

[0107] In view of this, in order to solve the problem that due to the extra space occupied by the protruding flange structure, the space utilization rate of the battery is low, and the energy density of the battery is affected, the applicant has conducted in-depth research and designed a battery. The battery includes a first box and a second box, the first box includes a first end wall and a first side wall, the second box includes a second end wall, the first end wall and the second end wall are disposed opposite to each other along a first direction, the second end wall has a first side surface in a second direction, and the first direction intersects with the second direction. A first side wall of the first box is connected to the first side surface of the second end wall.

[0108] In the above solutions, the flange structure protruding outward from the battery is eliminated, so that the first side wall is connected to the first side surface, saving space wasted by the flange structure, improving the space utilization rate of the battery in the second direction, accommodating more battery cells or reducing the volume of the battery, thus increasing the volumetric energy density of the battery.

[0109] The battery disclosed in the embodiments of this application may be applicable to, but is not limited to, use in a battery cabinet, a container-type energy storage device, etc. The energy storage device may include a plurality of batteries disclosed in this application.

[0110] The battery disclosed in the embodiments of this application may be applicable to, but is not limited to, use in an electrical device such as a vehicle, a ship or an aircraft. A power system of the electrical device may be composed of the battery disclosed in this application.

[0111] The embodiments of this application provide an electrical device using a battery as a power supply. The electrical device may be, but is not limited to, a mobile phone, a tablet computer, a laptop computer, an electric toy, an electric tool, an electric bicycle, an electric motorcycle, an electric vehicle, a ship, a heavy truck, a bus, a spacecraft, etc. The electric toy may include a stationary or mobile electric toy, e.g. a game console, an electric vehicle toy, an electric ship toy, and an electric airplane toy, etc. The spacecraft may include an airplane, a rocket, a space shuttle, and a spaceship, etc.

[0112] For ease of description in the following embodiments, a vehicle **1000** is used as an example of the electrical device according to an embodiment of this application.

[0113] Referring to FIG. **1**, FIG. **1** is a schematic diagram of a vehicle in some embodiments of this application. The vehicle **1000** may be an oil-fueled vehicle, a natural gas vehicle or a new energy vehicle, and the new energy vehicle may be a battery electric vehicle, a hybrid electric vehicle or a range-extended electric vehicle, etc. The type of the vehicle **1000** may be a car, an off-road vehicle, a heavy truck, or bus, etc. The interior of the vehicle **1000** is provided with a battery **100**, and the battery **100** may be disposed at the bottom, head or rear of the vehicle **1000**. The battery **100** may be configured to supply power to the vehicle **1000**. For example, the battery **100** may serve as an operating power source of the vehicle **1000** for circuitry of the vehicle **1000**, for example, for the working power requirements of the vehicle **1000** in start-up, navigation and operation.

[0114] The vehicle **1000** may further include a controller **200** and a motor **300**. The controller **200** is configured to control the battery **100** to supply power to the motor **300**, for example, to start or navigate the vehicle **1000**, or meet the operating power requirements of the vehicle in operation.

[0115] In some embodiments of this application, the battery **100** serves not only as an operating power supply of the vehicle **1000**, but may also serve as a drive power supply of the vehicle **1000** to provide a driving power supply for the vehicle **1000** in place of or partly in place of fuel oil or natural gas.

[0116] According to some embodiments of this application, this application provides a battery **100**. Please refer to FIGS. **2** to **6**. FIG. **2** is a schematic diagram of the battery in some embodiments of this application, FIG. **3** is a stereoscopic exploded diagram of the battery **100** in some embodiments of this application, FIG. **4** is a schematic diagram of a first box **10** in some embodiments of this application, FIG. **5** is a schematic diagram of a second box **20** in some embodiments of this application, and FIG. **6** is an enlarged view of A in FIG. **5**.

[0117] The battery **100** includes a battery cell (not shown in the figure), the first box **10** and the second box **20**. The first box **10** includes a first end wall **11** and a first side wall **12**. The second box **20** and the first box **10** are connected to each other to jointly enclose and form a closed space for accommodating the battery cell, the second box **20** includes a second end wall **21**, the second end wall **21** is disposed opposite to the first end wall **11** along a first direction z, the second box **20** has a first side surface **210** in a second direction x, and the first direction z intersects with the second direction x. One end of the first side wall **12** is connected to the first end wall **11**, and the other end of the first side wall **12** is connected to the first side surface **210**.

[0118] In some embodiments, the first box **10** may be regarded as an upper box of the battery **100**, the first end wall **11** may be regarded as a top wall of the battery **100**, and the first side wall **12** may be a portion connected to the first end wall **11** at one end and not in the same plane as the first end wall **11**; and the first side wall **12** may be connected to the first end wall **11** by welding, bonding or bolt connection, etc., and the first side wall **12** may also be integrally formed with the first end wall **11**.

[0119] The second box **20** may be regarded as a lower box of the battery **100**, the second end wall **21** may be regarded as a bottom wall of the battery **100**, and the first direction z may be a height direction of the battery **100**, that is, the top wall and the bottom wall are disposed opposite to each other along the height direction of the battery **100**. In some embodiments, the first side surface **210** may be a surface of the second end wall **21** in the second direction x. In some embodiments, the first side surface **210** may not be a surface of the second end wall **21** in the second direction x. For example, the first side surface **210** may be a surface of other components disposed on the second end wall **21** in the second direction x. In some embodiments, the first side surface **210** may protrude from a surface of the second end wall **21** in the second direction x towards the first box **10**.

[0120] The first box **10** and the second box **20** are connected to each other to form a closed space capable of accommodating battery cells.

[0121] The first direction z intersects with the second direction x, which may mean that the first direction z and the second direction x are not parallel. Correspondingly, the first side surface **210** may be understood as a surface of the second end wall **21** that does not face or face away from the first end wall **11**. In some embodiments, the first direction z is perpendicular to the second direction x.

[0122] "One end of the first side wall **12** is connected to the first end wall **11**, the other end of the first side wall is connected to the first side surface **210**" may mean that the first side wall **12** connects the first end wall **11** and the second end wall **21** to each other, and one end of the first side wall **12** away from the first end wall **11** is directly or indirectly connected to the first side surface **210**, which may be understood to mean that a connecting surface of the first side wall **12** and the first end wall **11** is located on the first side surface **210** and does not protrude from outer contours of the first box **10** and the second box **20** in the second direction x. "One end of the first side wall **12** away from the first end wall **11** is directly or indirectly connected to the first side surface **210**" may mean that the first side wall **12** is directly connected to the first side surface **210**, or the first side wall **12** is indirectly connected to the first side surface **210** through an intermediate connector.

[0123] In the above solutions, by connecting one end of the first side wall **12** of the first box **10** to the first side surface **210** of the second box **20** to achieve connection between the first box **10** and the second box **20**, the connection between the first box **10** and the second box **20** can be achieved without disposing a flange structure protruding along the second direction x, thereby improving the space utilization rate of the battery **100** in the second direction x to accommodate more battery cells or reduce the volume of the battery, and thus improving the volumetric energy density of the battery **100**.

[0124] According to some embodiments of this application, an inner surface of the first side wall **12** is connected to the first side surface **210**.

[0125] The inner surface of the first side wall **12** may refer to a surface of the first side wall **12** facing an interior of the battery **100**. "An inner surface of the first side wall **12** is connected to the first side surface **210**" may mean that a connection relationship between the inner surface of the first side wall **12** and the first side surface **210** is a face-to-face connection.

[0126] In the above solutions, by connecting the inner surface of the first side wall **12**, i.e., a surface facing the first side surface **210**, to the first side surface **210**, a size of the battery **100** in the second direction x can be effectively reduced to increase the volumetric energy density of the battery **100** as much as possible.

[0127] According to some embodiments of this application, referring to FIGS. **5** and **6**, the second end wall **21** has a first surface **211** facing the first end wall **11**, and the end that is of the first side wall **12** and that is away from the first end wall **11** exceeds the first surface **211** along a direction of the first end wall **11** pointing towards the second end wall **21**.

[0128] The second end wall **21** has the first surface **211** in the first direction z, and the first surface **211** is a surface of the second end wall **21** facing the first end wall **11**. The first surface **211** may be regarded as an upper surface of the second end wall **21**. In some embodiments, the battery cell is located between the first surface **211** and the first end wall **11**.

[0129] "The end that is of the first side wall **12** and that is away from the first end wall **11** exceeds the first surface **211** along a direction of the first end wall **11** pointing towards the second end wall **21**" may mean that an orthographic projection of the first side wall **12** on the plane where the first side surface **210** is located overlaps the first side surface **210**, and the portion where the first side wall **12** and the first side surface **210** overlap may be a portion where the first side wall **12** and the first side surface **210** are connected to each other.

[0130] In the above solutions, one end that is of the first side wall **12** and that is away from the first end wall **11** exceeds the first surface **211**, so that the first side wall **12** may be directly connected to the first side surface **210**, and on the one hand, the assembly difficulty and the manufacturing cost increase caused by the indirect connection of the first side wall **12** to the first side surface **210** through an intermediate connector can be effectively reduced, and on the other hand, an inner surface of a portion of the first end wall **11** exceeding the first surface **211** may be connected to the first side surface **210**, and a size of the battery **100** in the second direction x can be effectively reduced to increase the volumetric energy density of the battery **100** as much as possible.

[0131] According to some embodiments of this application, referring to FIGS. **5** and **6**, the second end wall **21** has a second surface **212** facing away from the first end wall **11**, and the end that is of the first side wall **12** and that is away from the first end wall **11** does not exceed the second surface **212** along a direction of the first end wall **11** pointing towards the second end wall **21**.

[0132] The second end wall **21** has the second surface **212** in the first direction z, and the second surface **212** is a surface that is of the second end wall **21** and that is facing away from the first end wall **11**. The second surface **212** may be regarded as a lower surface of the second end wall **21**.

[0133] "The end that is of the first side wall **12** and that is away from the first end wall **11** does not exceed the second surface **212** along a direction of the first end wall **11** pointing towards the second end wall **21**" may mean that the end that is of the first side wall **12** and that is away from the first end wall **11** is flush with the second surface **212** or is located in a position between the second surface **212** and the first end wall **11**.

[0134] In the above solutions, the end that is of the first side wall **12** and that is away from the first end wall **11** does not exceed the second surface **212**, so that a size of the battery **100** in the first direction z is controlled, and enabling the battery **100** to have a higher volumetric energy density.

[0135] In some embodiments, as shown in FIG. **3**, the second direction x is a width direction of the battery **100**. The battery **100** may be a prismatic battery. A height direction of the battery **100** may be the first direction z. The width direction and a length direction of the battery **100** may be directions which are perpendicular to each other, and both are perpendicular to the height direction. A size of the battery **100** in the length direction is generally larger than its size in the width direction, and in some embodiments, a side surface of the battery **100** in the width direction is a surface of the battery **100** with a larger area, i.e., a large surface. The first box **10** may be an upper box of the battery **100**, and in the manufacturing process of the battery **100**, the material cost and density of the upper box may be lower than those of the lower box. Therefore, when the second direction x is the width direction of the battery **100**, the first side wall **12** may be regarded as a large surface (a surface with a larger area) of the battery **100**, and the first side wall **12** may be disposed as large as possible, so that a portion of the second box **20** corresponding to the first side wall **12** is disposed as small as possible, thereby reducing the cost of the second box **20** and also reducing the overall weight of the battery **100**, so that the battery **100** is more lightweight, which is conducive to the improvement of the weight energy density of the battery **100**.

[0136] According to some embodiments of this application, a material density of the first box **10** is lower than a material density of the second box **20**.

[0137] The material density is the mass per unit volume of a material in a specific volumetric state.

[0138] In the above solutions, since the first side wall **12** is connected to the first side surface **210**, the proportion of the first box **10** to the battery **100** is increased, and the material density of the first box **10** is smaller than that of the second box **20**, the density of the battery **100** is reduced, and the weight of the battery **100** is reduced under the same volume, thus increasing the weight energy density of the battery.

[0139] In some other embodiments, the material density of the first box **10** may also be equal to or greater than the material density of the second box **20**.

[0140] According to some embodiments of this application, the material of the first box **10** is plastic, and the material of the second box **20** is aluminum alloy.

[0141] The cost of plastic is lower than the cost of aluminum alloy, and the density of plastic is lower than that of aluminum alloy. The structural strength and rigidity of aluminum alloy are higher than those of plastic.

[0142] In the above solutions, the first box **10** may be an upper box of the battery **100**, the second box **20** may be a lower box of the battery **100**, and the second box **20** is made of an aluminum alloy material, so that the second box **20** has higher structural strength and rigidity, and plays a strong protective role for the battery cell. As an upper box, the first box **10** plays a role of closing the battery cell between the upper and lower boxes, and it is made of plastic, so that the manufacturing cost of the battery **100** can be effectively reduced, and the weight of the battery **100** can be effectively reduced to improve the weight energy density of the battery **100**.

[0143] In some other embodiments, the first box **10** may also be made of materials such as aluminum, aluminum alloy, steel or stainless steel. In some other embodiments, the second box **20** may also be made of materials such as plastic, aluminum, steel, or stainless steel.

[0144] According to some embodiments of this application, the battery **100** further includes a first fastener (not shown in the figure), and the first side wall **12** is connected to the first side surface **210** through the first fastener.

[0145] The first fastener is a connecting component that connects the first side wall **12** and the first side surface **210**. In some embodiments, the first fastener may be a connecting piece such as a rivet, a screw or a bolt. In some other embodiments, the first fastener may also be an bonding layer disposed between the first side wall **12** and the first side surface **210**.

[0146] In the above solutions, the first fastener is disposed to connect the first side wall **12** and the first side surface **210** to improve the connection stability of the first side wall **12** and the first side surface **210**, the structural stability of the battery **100** is improved.

[0147] According to some embodiments of this application, in conjunction with FIGS. **6** and **7**, where FIG. **7** is a partial schematic diagram of a first box **10** in some embodiments of this application, the first side wall **12** is provided with a first through hole **120**, the first side surface **210** is provided with a first threaded hole **2100**, and the first fastener passes through the first through hole **120** and is connected to the first threaded hole **2100**.

[0148] The first through hole **120** may refer to a hole-shaped structure that penetrates an outer surface and an inner surface of the first side wall **12**. "The first side surface **210** is provided with a first threaded hole **2100**" may refer to a hole-shaped structure with an internal thread formed on the first side surface **210**, such as a self-tapping thread; or it may refer to a threaded hole of a blind rivet nut or a threaded hole of a pressing rivet nut disposed on the first side surface **210**. In conjunction with FIGS. **6** and **7**, the quantity of first through holes **120** may be plural, and a plurality of first through holes **120** are arranged at intervals along an extension direction of the first side wall **12** (the extension direction of the first side wall **12** is perpendicular to the first direction z and the second direction x); and the quantity of first threaded holes **2100** is also plural, and the first threaded holes **2100** are disposed corresponding to the first through holes **120**.

[0149] "The first fastener passes through the first through hole **120** and is connected to the first threaded hole **2100**" may mean that the first fastener has an external thread that mates with the first threaded hole **2100** to be capable of being threadedly connected with the first threaded hole **2100**.

[0150] In the above solutions, the first fastener may be a connecting component with the external thread such as a bolt and a screw which can be effectively connected to the first threaded hole **2100**, so that the connection stability between the first side wall **12** and the first side surface **210** is higher and the hermeticity between the first side wall **12** and the first side surface **210** is good.

[0151] According to some embodiments of this application, as shown in FIG. **3**, the battery **100** further includes a first sealing element **30**, the first sealing element **30** is disposed between the first side wall **12** and the first side surface **210**.

[0152] The first sealing element **30** may be a component with a sealing characteristic and disposed between the first side wall **12** and the first side surface **210**. In some embodiments, the first sealing element **30** may be a sealant or a sealing gasket, and the first sealing element **30** is clamped between the first side wall **12** and the first side surface **210**.

[0153] In the above solutions, by disposing the first sealing element **30** between the first side wall **12** and the first side surface **210**, the hermeticity between the first side wall **12** and the first side surface **210** can be improved, thus improving the hermeticity of the battery **100**.

[0154] According to some embodiments of this application, as shown in FIGS. **3**, **4** and **7**, a quantity of the first side walls **12** is two, and the two first side walls **12** are disposed opposite to each other along the second direction x. The second end wall **21** is located between the two first side walls **12**, and the second end wall **21** has two first side surfaces **210** disposed opposite to each other along the second direction x. The first side surface **210** corresponds one-to-one to the first side wall **12**.

[0155] In some embodiments, the quantity of the first side walls **12** is two, and the two first side walls **12** are disposed opposite to each other along the second direction x to correspond to the two first side surfaces **210** of the second end wall **21** disposed opposite to each other in the second direction x. The second end wall **21** is located between the two first side walls **12**, which may be understood as an outer contour of the battery **100** in the second direction x being defined by the two first side walls **12**.

[0156] In the above solutions, the quantity of the first side walls **12** is two, and the two first side walls **12** are disposed opposite to each other along the second direction x and are respectively connected to corresponding first side surfaces **210**, which can effectively improve the space utilization rate of the battery **100** in the second direction x, thus increasing the volumetric energy density of the battery **100**.

[0157] In some other embodiments, the first side walls **12** may be one, the one first side wall **12** can save space on one side of the second direction x for the battery **100**.

[0158] According to some embodiments of this application, as shown in FIG. **6**, the second box **20** further includes two second side walls **22**, the two second side walls **22** are disposed opposite to each other along the second direction x and connected to the second end wall **21**, and the battery cell is disposed between the two second side walls **22**.

[0159] The second side wall **22** is a component disposed on the second end wall **21**. The second side wall **22** may be disposed on the second end wall **21** by welding, bonding or bolt connection, etc., and the second side wall **22** may also be integrally formed with the second end wall **21**. The second end wall **21** has a first surface **211** facing the first end wall **11**, and the second side wall **22** protrudes from the first surface **211** along a direction of the second end wall **21** pointing towards the first end wall **11**.

[0160] In some embodiments, the first side surface **210** may be a surface where the second side wall **22** and the first side wall **12** are connected to each other.

[0161] In some embodiments, an outer surface of the second side wall **22** may be flush with the first side surface **210**.

[0162] In the above solutions, two opposite second side walls **22** are disposed on the second end wall **21** along the second direction x. On the one hand, the structural strength of the second box **20** can be improved, on the other hand, a mounting region of the battery cell can be defined, so that the battery cell can be orderly assembled in the second box **20**, and on the other hand, since the second side wall **22** protrudes from the surface of the second end wall **21**, when the battery cell is bonded to the second box **20** by glue, a glue overflow space can be effectively controlled, and the risk of waste and environmental pollution caused by glue overflow can be reduced.

[0163] According to some embodiments of this application, referring to FIGS. **4** and **7**, openings **13** are respectively formed at two ends of the first box **10** along the third direction y. Referring to FIG. **5**, the second box **20** further includes two third side walls **23**, the two third side walls **23** are disposed opposite to each other along the third direction y and connected to the second end wall **21**, and the two third side walls **23** respectively close the two openings **13**. The first direction z, the second direction x and the third direction y intersect in pairs. In some embodiments, the first direction z, the second direction x and the third direction y are perpendicular in pairs.

[0164] When the first direction z is a height direction of battery **100**, and the second direction x is a width direction of battery **100**, the third direction y may be a length direction of battery **100**.

[0165] The third side wall **23** is a component disposed on the second end wall **21**. The third side wall **23** may be connected to the second end wall **21** by welding, bonding or bolt connection, etc., and the third side wall **23** may also be integrally formed with the second end wall **21**. The second end wall **21** has a first surface **211** facing the first end wall **11**, and the third side wall **23** protrudes from the first surface **211** along a direction of the second end wall **21** pointing towards the first end wall **11**. The third side wall **23** may improve the structural strength of the first box **10**. In some embodiments, the third side wall **23** may be mounted with components such as explosion-proof valves, water-cooled connectors, or high- and low-voltage connectors to enable the battery **100** to perform normal charging and discharging operations.

[0166] The openings **13** respectively formed at two ends of the first box **10** along the third direction y may be notch-shaped portions corresponding to the third side wall **23**, and the third side wall **23** can correspondingly close the openings **13**, avoiding generating additional mutual interference or overlapping portions in the battery **100**, thereby improving the utilization rate of manufacturing materials of the battery **100**, and enabling the battery **100** to have a lower manufacturing cost; and at the same time, it also avoids the occurrence of generating additional mutual interference or overlapping portions in the battery **100** to cause space waste. In some embodiments, a shape of a cross-section of the first box **10** may be a U shape.

[0167] In the above solutions, by disposing a third side wall **23**, on the one hand, components such as explosion-proof valves, water-cooled connectors, or high- and low-voltage plug-in connectors can be mounted on the third side wall **23** to enable the battery **100** to perform normal charging and discharging operations; and on the other hand, compared with a flange structure protruding along the third direction y disposed between the first box **10** and the second box **20**, by disposing the third side wall **23** to close an opening **13**, the space utilization rate of the battery **100** in the third direction y can be improved, and the volumetric energy density of the battery **100** can be increased.

[0168] According to some embodiments of this application, referring to FIGS. **7** and **8**, where FIG. **8** is a schematic diagram of a third side wall **23** in some embodiments of this application, the third side wall **23** includes a third surface **230** facing the closed space, a fourth surface **231** facing away from the closed space and a second side surface **232** connecting the third surface **230** and the fourth surface **231**; and the first box **10** includes two connecting portions **14**, the two connecting portions **14** are respectively located at the two ends of the first box **10** along the third direction y, the connecting portion **14** peripherally forms the opening **13**, and the connecting portion **14** is connected to the second side surface **232**.

[0169] The third surface **230** may be an inner surface of the third side wall **23**, the fourth surface **231** may be an outer surface of the third side wall **23**, and the second side surface **232** may be an outer peripheral surface of the third side wall **23** located between the inner surface and the outer surface. The outer peripheral surface of the third side wall **23** may refer to an outer surface of the third side wall **23** in the first direction z and the second direction x.

[0170] The connecting portion **14** is a component located at an end portion of the first box **10** in the third direction y, which peripherally forms the opening **13** of the first box **10**, and a contour of the connecting portion **14** fits with the second side surface **232** of the third side wall **23**.

[0171] "The connecting portion **14** is connected to the second side surface **232**" may mean that the first box **10** and the second box **20** are connected to each other by the connecting portion **14** and the second side surface **232** in the third direction y.

[0172] Referring to FIG. **7**, the connecting portion **14** may be sheet-shaped, which mates with the second side surface **232**. In some embodiments, a connection relationship between the connecting portion **14** and the second side surface **232** is a face-to-face connection.

[0173] In the above solutions, by disposing a connecting portion **14** to be connected to the second side surface **232** of the third side wall **23**, the connection stability and hermeticity between the first box **10** and the second box **20** can be improved.

[0174] According to some embodiments of this application, referring to FIG. **8**, the second side surface **232** includes a first flat surface **2320**, a second flat surface **2321** and a transition surface **2322**. The first flat surface **2320** is disposed at one end that is of the third side wall **23** and that is facing away from the second end wall **21**, and two second flat surfaces **2321** are respectively disposed at two ends of the third side wall **23** along the second direction x. The transition surface **2322** connects the first flat surface **2320** and the second flat surface **2321**, so that the first flat surface **2320** smoothly transitions to the second flat surface **2321**.

[0175] The first flat surface **2320** may be an upper surface of the third side wall **23**, which may be a flat surface. The second flat surface **2321** may be an outer side surface of the third side wall **23**, which may be a smooth surface. In some embodiments, the first flat surface **2320** and the second flat surface **2321** are perpendicular to each other. The transition surface **2322** is a portion connecting the first flat surface **2320** and the second flat surface **2321**, which enables a smooth transition between the first flat surface **2320** and the second flat surface **2321**. The smooth transition may refer to a non-right angle transition relationship between the first flat surface **2320** and the second flat surface **2321**.

[0176] In some embodiments, the transition surface **2322** may be an arc surface or a beveled surface.

[0177] In the above solutions, the first flat surface **2320** and the second flat surface **2321** are non-coplanar surfaces. Therefore, by disposing the transition surface **2322**, a smooth transition between the first flat surface **2320** and the second flat surface **2321** is allowed, which can be conducive to forming a good sealing surface between the connecting portion **14** and the second side surface **232**, improving the hermeticity of the battery **100**.

[0178] According to some embodiments of this application, reference is made to FIG. **9**, where FIG. **9** is a schematic diagram of a first flat surface, a second flat surface and a transition surface in some embodiments of this application.

[0179] The transition surface **2322** is a circular arc transition surface.

[0180] The circular arc transition surface may refer to a surface extending along a circular arc-shaped trajectory. The transition surface **2322** is a circular arc transition surface, which may be understood to mean that the first flat surface **2320** and the second flat surface **2321** are connected by a circular arc surface, that is, the first flat surface **2320** and the second flat surface **2321** may have no edges and corners.

[0181] In some embodiments, a sealing structure may be disposed between the first box **10** and the second box **20**, so that the first box **10** is in hermetic connection with the second box **20**. For example, a second sealing element **31** (see FIG. **3**) may be disposed between the second side surface **232** and the connecting portion **14**, and the second sealing element **31** may be supported by the transition surface **2322**. Since the first flat surface **2320** transitions to the second flat surface **2321** through the circular arc surface, the risk of stress concentration of the second sealing element **31** is small.

[0182] In the above solutions, by disposing the transition surface **2322** as a circular arc transition surface, the first flat surface **2320** can smoothly transition to the second flat surface **2321**, so that the sealing structure can be smoothly disposed between the third side wall **23** and the first box **10**, reducing the risk of sealing failure due to damage to the sealing structure caused by stress concentration of the sealing structure, which is conducive to the sealing performance between the second side surface **232** and the connecting portion **14**, and improving the hermeticity and reliability of the battery **100**.

[0183] According to some embodiments of this application, the transition surface **2322** has a first edge **23220** in the third direction y, the first edge **23220** is circular arc-shaped, and a radius of the first edge **23220** is greater than or equal to 10 mm.

[0184] The first edge **23220** may correspond to an extension trajectory of the transition surface **2322**, and the first edge **23220** may also refer to an edge of the transition surface **2322** parallel to its extension direction. In some embodiments, one end of the first edge **23220** is connected to the first flat surface **2320**, and the other end of the first edge is connected to the second flat surface **2321**.

[0185] In some embodiments, the first edge **23220** may be a circular arc line, and the radius of the first edge **23220** is greater than or equal to 10 mm. For example, the first edge **23220** may be 10 mm, 11 mm, 12 mm, 13 mm, or greater.

[0186] In some embodiments, rounding the first flat surface **2320** and the second flat surface **2321** can form a transition surface **2322** in the form of a circular arc surface.

[0187] In the above solutions, by defining the first edge **23220** of the transition surface as a circular arc shape, the transition surface extends in an arc-shaped trajectory, and by disposing the radius of the first edge **23220** to be greater than or equal to 10 mm, the first flat surface **2320** can smoothly transition to the second flat surface **2321**, and the sealing structure can be gently disposed between the third side wall **23** and the first box **10**, effectively reducing the risk of sealing failure due to damage to the sealing structure caused by stress concentration of the sealing structure, which is conducive to the sealing performance between the second side surface and the connecting portion, and improving the hermeticity and reliability of the battery.

[0188] According to some embodiments of this application, referring to FIG. **8**, the transition surface **2322** includes a first transition surface **2322*a*** and a second transition surface **2322*b*** connected to each other, an edge of the first transition surface **2322*a*** in the third direction y is smooth, and an edge of the second transition surface **2322*b*** in the third direction y is circular arc-shaped.

[0189] "The edge of the first transition surface **2322*a*** in the third direction y is smooth" may be understood to mean that the first transition surface **2322*a*** is a flat surface, and its surface is smooth and not a curved surface. "The edge of the second transition surface **2322*b*** in the third direction y is circular arc-shaped" may be understood to mean that the second transition surface **2322*b*** is a circular arc surface and its surface is a curved surface.

[0190] In some embodiments, the first transition surface **2322*a*** may be provided with a locking position for mating with a locking piece. For example, the first transition surface **2322*a*** may be provided with a second threaded hole **2323**, and the second threaded hole is configured to mate with the locking piece (e.g. a second fastener), so that the second side surface **232** is connected to the connecting portion **14**.

[0191] In the above solutions, by disposing the first transition surface **2322*a*** and the second transition surface **2322*b***, on the one hand, the first flat surface **2320** can smoothly transition to the second flat surface **2321**, which is conducive to improving the sealing performance between the second side surface **232** and the connecting portion **14**, and improving the hermeticity and reliability of the battery **100**; and on the other hand, the first transition surface **2322*a*** can provide a flat surface for locking of the third side wall **23** and the first box **10**, so that a locking piece (such as a bolt) can pass through the first box **10** and be locked into the first transition surface **2322*a*** stably and with good hermeticity, thereby improving the structural stability of the battery **100**, and enabling the battery **100** to have higher reliability.

[0192] According to some embodiments of this application, a quantity of the first transition surfaces **2322*a*** is n, with n≥1, and a quantity of the second transition surfaces **2322*b*** is n+1; and any one of the first transition surfaces **2322*a*** is disposed between two adjacent second transition surfaces **2322*b***.

[0193] The quantity of second transition surfaces **2322*b*** is one more than the quantity of first transition surfaces **2322*a***, and any one of the first transition surfaces **2322*a*** is disposed between two adjacent second transition surfaces **2322*b***. It may be understood that the first transition surface **2322*a*** and the second transition surface **2322*b*** are alternately connected, and a portion where the transition surface **2322** is connected to the first flat surface **2320** and the second flat surface **2321** is the second transition surface **2322*b***.

[0194] In some embodiments, as shown in FIG. **8**, the quantity of first transition surfaces **2322*a*** is 1, and the quantity of second transition surfaces **2322*b*** is 2. The first transition surface **2322*a*** is located between two second transition surfaces **2322*b***.

[0195] In other embodiments, the quantity of first transition surfaces **2322*a*** may be 3, and when the quantity of first transition surfaces **2322*a*** is 3, the quantity of second transition surfaces **2322*b*** is 4. In other embodiments, when the quantity of first transition surfaces **2322*a*** may be other values, for example, 4, 5 or 6, the quantity of second transition surfaces **2322***b* is always one more than the quantity of the first transition surfaces **2322***a*, for example, 5, 6 or 7.

[0196] In the above solutions, by disposing the quantity of second transition surfaces **2322***b* to be one more than that of the first transition surfaces **2322***a* and disposing any one of the first transition surfaces **2322***a* between the two adjacent second transition surfaces **2322***b*, on the one hand, the first flat surface **2320** can smoothly transition to the second flat surface **2321**, that is, the transition surfaces **2322** connected to the first flat surface **2320** and the second flat surface **2321** are both the second transition surfaces **2322***b* with arc surfaces, reducing the risk of sealing failure caused by formation of edges and corners between the flat surfaces. On the other hand, by disposing the first transition surface **2322***a*, a plurality of locking positions can be provided between the third side wall **23** and the first box **10**, which is conducive to effective connection of the locking piece to the third side wall **23** and the first box **10**, improving the structural stability of the battery **100**, and enabling the battery to have higher reliability.

[0197] According to some embodiments of this application, a length of the first transition surface **2322***a* is greater than or equal to 6 mm.

[0198] The length of the first transition surface **2322***a* is L, and a length direction of the first transition surface **2322***a* may be perpendicular to a thickness direction of the third side wall **23**.

[0199] A value of the length of the first transition surface **2322***a* may be 6 mm, 7 mm, 8 mm, or larger values.

[0200] In the above solutions, by disposing the length of the first transition surface **2322***a* to be greater than or equal to 6 mm, one or more locking positions can be disposed on the first transition surface **2322***a* (see FIG. **8**, two second threaded holes **2323** are disposed on the first transition surface **2322***a*), which is conducive to effective connection of the locking piece to the third side wall **23** and the first box **10**, improving the structural stability of the battery, and enabling the battery **100** to have higher reliability.

[0201] According to other embodiments of this application, reference is made to FIG. **10**, where FIG. **10** is a schematic diagram of a third side wall in some other embodiments of this application.

[0202] The second side surface **232** includes a circular arc surface **2324** and second flat surfaces **2321**. The two second flat surfaces **2321** are respectively disposed at the two ends of the third side wall **23** along the second direction x. One end of one of the two second flat surfaces **2321** and one end of the other of the two second flat surfaces **2321** face away from each other, and are connected by the circular arc surface **2324**.

[0203] The circular arc surface **2324** may be an upper surface of the third side wall **23**, the circular arc surface **2324** may be a curved surface, and an outer edge of the curved surface may be circular arc-shaped. The second flat surface **2321** may be an outer side surface of the third side surface, which may be a smooth surface. A plurality of locking positions may be disposed on the second flat surface **2321**. For example, a plurality of second threaded holes **2323** may be disposed to connect the third side wall **23** to the first box **10**.

[0204] In the above solutions, a smooth transition between two second flat surfaces **2321** disposed opposite to each other through a circular arc surface **2324** can be conducive to forming a good sealing surface between the connecting portion **14** and the second side surface **232**, enabling the battery **100** to have higher hermeticity.

[0205] According to some embodiments of this application, referring to FIG. **8**, the second flat surface **2321** is flush with the first side surface **210**.

[0206] The second flat surface **2321** is flush with the first side surface **210**, which means that the outer side surface of the third side wall **23** in the second direction x is flush with the outer side surface of the second end wall **21** in the second direction x, that is, a surface of the second box **20** in the second direction x is a smooth surface.

[0207] In the above solutions, the second flat surface **2321** is flush with the first side surface **210**, which can reduce the risk of generating a gap between the first side wall **12** and the second side surface **232**, reduce the risk of generating a gap between the connecting portion **14** and the second flat surface **2321**, and thus improve the hermeticity of the battery **100**.

[0208] According to other embodiments of this application, reference is made to FIG. **11**, where FIG. **11** is a partial schematic diagram of a second box in some other embodiments of this application.

[0209] The second end wall **21** includes a first portion **213** and a second portion **214**, and the second portion **214** is located on one side of the first portion **213** and connected to the first portion **213** along the second direction x. A size of the first portion **213** in the second direction x is greater than a size of the second portion **214** in the second direction x.

[0210] The second end wall **21** may include a first portion **213** and a second portion **214**. The phrase "the second portion **214** is located on one side of the first portion **213** and connected to the first portion **213** along the second direction x" may refer to an end portion of the first portion **213** being connected to the second portion **214** in the third direction y. In some embodiments, two ends of the first portion **213** may be provided with a second portion **214** in the third direction y.

[0211] "A size of the first portion **213** in the second direction x is greater than a size of the second portion **214** in the second direction x" may be understood to mean that a size of an end portion of the second end wall **21** in the second direction x is reduced.

[0212] In some embodiments, the first portion **213** may be a portion of the second end wall **21** for carrying the battery cell, and the second portion **214** may not carry the battery cell. For example, a projection of the battery cell does not fall on the second portion **214** along the first direction z.

[0213] In the above solutions, the first end wall **21** includes a first portion **213** and a second portion **214**. By disposing the size of the first portion **213** in the second direction x to be greater than the size of the second portion **214** in the second direction x, a notch is formed between the first portion **213** and the second portion **214**, or in other words, an end portion of the second end wall **21** is narrowed. Since the end portion of the second end wall **21** is not utilized by the battery cell, by narrowing the portion, on the one hand (for example, the narrowed portion can form an avoidance space to avoid other members of an electrical device), the volume of the battery **100** can be reduced, and the mounting space of the battery **100** can be saved, and on the other hand, an effect of reducing the weight of the battery **100** can be achieved, so that the battery **100** is lightweight.

[0214] According to some embodiments of this application, the second portion **214** has a third side surface **2140** in the second direction x, and the third side surface **2140** is flush with the second flat surface **2321**.

[0215] The third side surface **2140** is a surface of the second portion **214** in the second direction x. The third side surface **2140** may be parallel to the first side surface **210**.

[0216] The third side surface **2140** is flush with the second flat surface **2321**, which may mean that a size of the third side wall **23** in the second direction x may be equal to a size of the second portion **214** in the second direction x.

[0217] In the above solutions, disposing the second flat surface **2321** to be flush with the third side surface **2140** of the second portion **214** is conducive to hermeticity between the third side surface **2140** and the second end wall **21** and the first box **10**, reducing the risk of sealing failure between the first box **10** and the second box **20** due to a gap generated by the third side wall **23** and the second end wall **21** at the second portion **214**, and enabling the battery **100** to have higher reliability.

[0218] According to some embodiments of this application, the first portion **213** has a fourth side surface **2130** in the second direction x, and the third side surface **2140** protrudes from the fourth side surface **2130** along the third direction y.

[0219] The fourth side surface **2130** is a surface of the first portion **213** in the second direction x. The fourth side surface **2130** may be parallel to the first side surface **210**.

[0220] The third side surface **2140** protrudes from the fourth side surface **2130** along the third direction y, which may be understood as the third side surface **2140** exceeding the fourth side surface **2130** in the third direction y.

[0221] In the above solutions, the third side surface **2140** protrudes from the fourth side surface **2130** along the third direction y, that is, the second flat surface **2321** that is flush with the third side surface **2140** also protrudes from the fourth side surface **2130**, enabling a notch to be formed at a portion of the first box **10** and the second box **20** corresponding to a portion between the third side surface **2140** and the fourth side surface **2130**, or it is understood that the first box **10** and the second box **20** are narrowed at the portion, which on the one hand, can reduce the volume of the battery **100** and save the mounting space of the battery **100**, and on the other hand, can achieve the effect of reducing the weight of the battery **100**, so that the battery is lightweight.

[0222] According to some embodiments of this application, the first portion **213** has a third flat surface **2131** in the third direction y, and one end of the third flat surface **2131** is in circular arc transition connection with the third side surface **2140**, and the other end of the third flat surface **2131** is in circular arc transition connection with the fourth side surface **2130** along the second direction x.

[0223] The third flat surface **2131** is a surface of the first portion **213** in the third direction y. The third flat surface **2131** may be parallel to the fourth surface **231**.

[0224] Rounding may be performed between the third flat surface **2131** and the third side surface **2140** to allow a circular arc transition connection between the third flat surface **2131** and the third side surface **2140**. Rounding may be performed between the third flat surface **2131** and the fourth side surface **2130** to allow a circular arc transition connection between the third flat surface **2131** and the fourth side surface **2130**.

[0225] In the above solutions, the third flat surface **2131** is disposed between the third side surface **2140** and the fourth side surface **2130**. By disposing one end of the third flat surface **2131** to be in circular arc transition connection with the third side surface **2140** and the other end to be in circular arc transition connection with the fourth side surface **2130**, the risk of damage to a sealing structure between the second end wall **21**, the third side wall **23** and the first box **10** due to stress concentration can be reduced, which is conducive to the hermeticity between the first box **10** and the second box **20**, and enabling the battery **100** to have higher reliability.

[0226] According to other embodiments of this application, reference is made to FIG. **12**, where FIG. **12** is a partial schematic structural diagram of a second box in some other embodiments of this application.

[0227] The second side surface **23** includes the first flat surface **2320** and the transition surface **2322**. The first flat surface **2320** is disposed at one end that is of the third side wall **23** and that is facing away from the second end wall **21**, and an end portion of the first flat surface **2320** smoothly transitions to a side surface of the second end wall **21** through the transition surface **2322**.

[0228] In some embodiments, the first flat surface **2320** may be an upper surface of the third side wall **23**, which may be a smooth surface. The transition surface **2322** may be an outer side surface of the third side wall **23**, which may be a curved surface. For example, the transition surface **2322** is a circular arc surface.

[0229] "An end portion of the first flat surface **2320** smoothly transitions to a side surface of the second end wall **21** through the transition surface **2322**" may be understood to mean that there is no edge and corner between the first flat surface **2320** and the side surface of the second end wall **21**, and the first flat surface **2320** and the second end wall **21** are connected by the transition surface **2322** with the circular arc surface.

[0230] In the above solutions, the second side surface **23** includes the first flat surface **2320** and the transition surface **2322**. The first flat surface **2320** can form a good seal with the connecting portion **14**. By disposing the transition surface **2322** between the first flat surface **2320** and a side surface of the second end wall **21**, a smooth transition between the first flat surface **2320** and the side surface of the second end wall **21** can be achieved, which can be conducive to forming a good sealing surface between the first box **10** and the second box **20**, enabling the battery **100** to have higher hermeticity.

[0231] According to some other embodiments of this application, reference is made to FIGS. **13** and **14**, where FIG. **13** is a partial schematic diagram of a first box in some other embodiments of this application, and FIG. **14** is a partial schematic diagram of a second box in some other embodiments of this application. The first box **10** further includes a fourth side wall **15**, and the fourth side wall **15** is adjacent to the first side wall **12**. One end of the fourth side wall **15** is connected to the first end wall **11**, the connecting portion **14** is disposed at the other end of the fourth side wall **15**, and the connecting portion **14** protrudes from the fourth side wall **15** in a direction facing away from the closed space.

[0232] The first end wall **11** has a lower surface facing the second end wall **21**, the fourth side wall **15** is a component protruding from the lower surface and adjacent to the first side wall **12**, the fourth side wall **15** may be connected to the first end wall **11** by welding, bonding or bolt connection, and the fourth side wall **15** may also be integrally formed with the first end wall **11**. When the first box **10** has two first side walls **12** disposed opposite to each other along the second direction x, the fourth side wall **15** is located between the two first side walls **12**, one end of the fourth side wall **15** is connected to the first end wall **11**, and two opposite ends of the fourth side wall **15** in the second direction x are respectively connected to the two first side walls **12**.

[0233] The connecting portion **14** may be protrusively disposed on the fourth side wall **15** from an interior of the closed space to an exterior of the closed space, the connecting portion **14** may be integrally formed with the fourth side wall **15**, or the connecting portion **14** may be connected to the fourth side wall **15** by welding, bonding or bolt connection, etc.

[0234] In some embodiments, in conjunction with FIGS. **13** and **14**, by disposing the fourth side wall **15**, a size of the third side wall **23** in the first direction z can be reduced, that is, the larger a size of the fourth side wall **15** in the first direction z, the smaller the size of the third side wall **23** in the first direction z.

[0235] In the above solutions, the first box **10** may be an upper box, and the second box **20** may be a lower box. In a manufacturing process of the battery **100**, the material cost and density of the upper box may be lower than those of the lower box. Therefore, by disposing a fourth side wall **15** to increase the proportion of the first box **10** in the battery **100**, the manufacturing cost of the battery **100** is effectively reduced and the weight energy density of the battery **100** is increased.

[0236] In some embodiments, a size of the fourth side wall **15** in the first direction z may be disposed larger, and components such as explosion-proof valves, water-cooled connectors, or high- and low-voltage plug-in connectors may be mounted on the fourth side wall **15**.

[0237] According to some embodiments of this application, the battery **100** further includes a second fastener (not shown in the figure), and the connecting portion **14** is connected to the second side surface **232** through the second fastener.

[0238] The second fastener is a connecting component that connects the second side surface **232** and the connecting portion **14**. In some embodiments, the second fastener may be a connecting piece such as a rivet, a screw and a bolt. In still other embodiments, the second fastener may also be an adhesive layer disposed between the connecting portion **14** and the second side surface **232**. Referring to FIG. **8**, when the second fastener is a threaded piece, such as a bolt, a corresponding second threaded hole **2323** may be disposed on the second side surface **232**.

[0239] In the above solutions, by disposing the second fastener to connect the second side surface **232** and the connecting portion **14**, the connection stability between the connecting portion **14** and the second side surface **232** is improved, and the structural stability of the battery **100** is improved.

[0240] According to some embodiments of this application, as shown in FIGS. **7** and **8**, the connecting portion **14** is provided with a second through hole **140**, the second side surface **232** is provided with the second threaded hole **2323**, and the second fastener passes through the second through hole **140** and is connected to the second threaded hole **2323**.

[0241] The second through hole **140** may refer to a hole-shaped structure that penetrates an outer surface and an inner surface of the connecting portion **14**. "The second side surface **232** is provided with the second threaded hole **2323**" may refer to a hole-shaped structure with an internal thread formed on the second side surface **232**, such as a self-tapping thread; or it may refer to a blind rivet nut or a pressing rivet nut disposed on the second side surface **232**. In conjunction with FIGS. **7** and **8**, the quantity of second through holes **140** may be plural, and a plurality of second through holes **140** may be arranged at intervals; the quantity of second threaded holes **2323** is also plural, and the first threaded holes **2100** is disposed corresponding to the first through holes **120**.

[0242] "The second fastener passes through the second through hole **140** and is connected to the second threaded hole **2323**" may mean that the second fastener has an external thread that mates with the second threaded hole **2323** to be capable of being threadedly connected with the second threaded hole **2323**.

[0243] In the above solutions, the second fastener may be a connecting component with an external thread such as a bolt and a screw, which can be effectively connected to the second threaded hole **2323**, improve the connection stability between the connecting portion **14** and the second side surface **232**, and allow good hermeticity between the connecting portion **14** and the second side surface **232**.

[0244] According to some embodiments of this application, as shown in FIG. **3**, the battery **100** further includes a second sealing element **31**, the second sealing element **31** is disposed between the connecting portion **14** and the second side surface **232**.

[0245] The second sealing element **31** may be a component with a sealing characteristic and disposed between the connecting portion **14** and the second side surface **232**. In some embodiments, the second sealing element **31** may be a sealant or a sealing gasket, and the second sealing element **31** is clamped between the first flat surface **2320** and the connecting portion **14**, between the second flat surface **2321** and the connecting portion **14**, and between the transition surface **2322** and the connecting portion **14**.

[0246] In the above solutions, by disposing the second sealing element **31** between the connecting portion **14** and the second side surface **232**, the hermeticity between a wall of the connecting portion **14** and the second side surface **232** can be improved, thus improving the hermeticity of the battery **100**.

[0247] According to some embodiments of this application, as shown in FIG. **3**, the battery **100** further includes a first sealing element **30**, the first sealing element **30** is disposed between the first side wall **12** and the first side surface **210**. Two ends of the first sealing element **30** are respectively connected to two second sealing elements **31**.

[0248] In FIG. **3**, the first sealing element **30** corresponds to a connecting portion **14** between the first side surface **210** and the first side wall **12**, which may extend along the length direction of the battery **100**. The second sealing element **31** corresponds to the connecting part between the second side surface **232** and the connecting portion **14**, which includes a portion extending along the width direction of the battery **100**, a portion extending along the height direction of the battery **100** and a portion corresponding to the transition surface **2322**.

[0249] In some embodiments, the first sealing element **30** and the second sealing element **31** may be structures independent of each other, and the first sealing element **30** and the second sealing element **31** are connected by bonding or in an intermediate connector. In other embodiments, the first sealing element **30** and the second sealing element **31** may be an integrated structure.

[0250] In the above solutions, the first sealing element **30** and the second sealing element **31** may be integrally formed or separately connected. By disposing the first sealing element **30** and the second sealing element **31**, the good hermeticity between the first box **10** and the second box **20** can be achieved, thereby improving the safety of the battery.

[0251] According to still other embodiments of this application, referring to FIG. **15**, where FIG. **15** is a schematic diagram of a second box in still other embodiments of this application, a size of one of the third side walls **23** along the first direction z is smaller than a size of the other one of the third side walls **23** along the first direction z.

[0252] The size of the third side wall **23** in the first direction z may be regarded as a height of the third side wall **23**. "A size of one of the third side walls **23** along the first direction z is smaller than a size of the other one of the third side walls **23** along the first direction z" may mean that the one of the third side walls **23** is higher and the other one of the third side walls **23** is lower. Since the heights of the two third side walls **23** are different, the resulting height difference may be compensated by the first box **10**.

[0253] In the above solutions, the larger third side wall **23** may be mounted with components such as explosion-proof valves, water-cooled connectors, or high- and low-voltage plug-in connectors to enable the battery **100** to perform normal charging and discharging operations; and by reducing the size of the other one of the third side walls **23**, the size of a portion of the first box **10** corresponding to the third side wall **23** can be adaptively increased, increasing the proportion of the first box **10** to the battery **100** (in this embodiment, the first box **10** may be an upper box with a lower material cost and density), thereby lowering the manufacturing cost of the battery **100** and increasing the weight energy density of the battery **100**.

[0254] According to other embodiments of this application, reference is made to FIG. **16**, where FIG. **16** is a schematic diagram of a first box and a second box in some embodiments of this application.

[0255] The first box **10** has a fifth side surface **16** in the third direction y, the second box **20** has a sixth side surface **25** in the third direction y, the sixth side surface **25** is disposed adjacent to the first side surface **210**, the fifth side surface **16** and the sixth side surface **25** are connected to each other, and the first direction z, the second direction x and the third direction y intersect in pairs.

[0256] The fifth side surface **16** may be a surface of the first box **10** in the third direction y, and the fifth side surface **16** is configured to be connected to the sixth side surface **25** of the second box in the third direction y. In some embodiments, the fifth side surface **16** protrudes from the first end wall **11** along the first direction z and extends towards the second end wall **21**.

[0257] The sixth side surface **25** may be a surface of the second box **20** in the third direction y, and the sixth side wall **25** is configured to be connected to the fifth side surface **16** of the first box **10** in the third direction y. In some embodiments, the sixth side surface **25** protrudes from the first surface **211** along the first direction z and extends towards the first end wall **11**. In some embodiments, the sixth side surface **25** may not exceed the first surface.

[0258] In some embodiments, the first box **10** may be regarded as a semi-closed structure, which has an opening facing the second box **20** in the first direction z, the second box **20** is connected to the first side wall **12** through the first side surface **210**, and the opening can be closed by connecting the sixth side surface **25** to the fifth side surface **16**.

[0259] In the above solutions, by disposing the sixth side surface **25** to be connected to the fifth side surface **16**, the connection between the first box **10** and the second box **20** can be achieved without disposing a flange structure protruding along the third direction y, thereby improving the space utilization rate of the battery **100** in the third direction y to accommodate more battery cells or reducing the volume of the battery, and thus improving the volumetric energy density of the battery.

[0260] According to some embodiments of this application, referring to FIG. **16**, the sixth side surface **25** is in a circular arc transition to the first side surface **210**.

[0261] Rounding may be performed between the sixth side surface **25** and the first side surface **210** to allow connection between the sixth side surface **25** and the first side surface **210** through an arc surface.

[0262] In the above solutions, disposing the sixth side surface **25** to be in a circular arc transition to the first side surface **210** can be conducive to forming a good sealing surface between the second box **20** and the first box **10**, reducing the risk of damage to a sealing structure disposed between the second box **20** and the first box **10** due to stress concentration, and enabling the battery **100** to have higher hermeticity and reliability.

[0263] According to some embodiments of this application, as shown in FIGS. **3** and **5**, the second end wall **21** is provided with a mounting portion **40** for mounting the battery **100** to an electrical device.

[0264] The mounting portion **40** is a component disposed on the second end wall **21** for mounting the battery **100** to the electrical device. The mounting portion **40** may be a connection structure such as a nut or a connecting bracket disposed on the second end wall **21**. In some embodiments, the quantity of mounting portions **40** may be plural, which may be arranged along the third direction y and the second direction x. In some embodiments, the mounting portion **40** may be M8, M10, M12, or M16 threaded hole structures provided on the second end wall **21**.

[0265] In the above solutions, the second end wall **21** is provided with the mounting portion **40** to achieve stable assembly of the battery **100**, so that the battery **100** can provide stable electrical energy.

[0266] According to some embodiments of this application, an electrical device is further provided. The electrical device includes the battery **100** described above, and the battery **100** is configured to provide electrical energy. In some embodiments, the electrical device may be a vehicle, and the type of the vehicle may be a heavy truck or a bus.

[0267] According to some embodiments of this application, this application further provides a battery **100**, as shown in FIGS. **3-8**. The battery **100** includes a first box **10**, a second box **20** and a battery cell. The first box **10** is an upper box of the battery **100**, the second box **20** is a lower box of the battery **100**, the second box **20** and the first box **10** are connected to each other to jointly enclose and form a closed space, and the battery cell is disposed in the closed space. The first box **10** includes a first end wall **11**, two first side walls **12** and two connecting portions **14**. The first end wall **11** may be a top wall of the battery **100**, and the two first side walls **12** are disposed opposite to the first end wall **11** along a second direction x (a width direction of the battery **100**). The two connecting portions **14** are respectively located at two ends of the first box **10** in a third direction y (a length direction of the battery **100**), and the connecting portions **14** peripherally form an opening **13**. The second box **20** includes a second end wall **21** and two second side walls **22**. The second end wall **21** may be a bottom wall of the battery **100**, and the second box **20** has two first surfaces **211** disposed opposite to each other along the second direction x. The second end wall **21** may be provided with crossbeams **24**, and the quantity of crossbeams **24** may be plural. A plurality of crossbeams **24** are arranged at intervals along the third direction y. The crossbeams **24** may improve the structural strength of the second box **20**, and the battery cell may also be connected to the crossbeams **24**. The two second side walls **22** are disposed opposite to each other on the second end wall **21** along the third direction y (the length direction of battery **100**). The second side wall **22** has a first flat surface **2320** facing the first end wall **11**, two second flat surfaces **2321** disposed opposite to each other along the second direction x, and a transition surface **2322** connecting the first flat surface **2320** and the second flat surfaces **2321**.

[0268] The first end wall **11** and the second end wall **21** are disposed opposite to each other along the first direction z (a height direction of the battery **100**). An inner surface of the first side wall **12** is connected to the first side surface **210**, and a first sealing element **30** is disposed between the first side wall **12** and the first side surface **210**. The first side wall **12** and the second side surface **232** are connected by a first fastener. The second side wall **22** corresponds to the connecting portion **14**, and a second sealing element **31** is disposed between the second side wall **22** and the connecting portion **14**. The first flat surface **2320**, the second flat surface **2321**, and the transition surface **2322** are all connected to the connecting portion **14** through a second fastener.

[0269] The crossbeam **24** is provided with a mounting portion **40** for mounting the battery **100** to an electrical device (such as a heavy truck or a bus).

[0270] The foregoing are merely optional embodiments of this application and are not intended to limit this application. To a person skilled in the art, various modifications and variations may be made to this application. Any modifications, equivalent substitutions, improvements, etc. made within the spirit and principles of this application shall be included within the scope of protection of this application.

What is claimed is:

1. A battery, wherein the battery comprises:

a battery cell;

a first box, wherein the first box comprises a first end wall and a first side wall; and

a second box, wherein the second box and the first box are connected to each other to jointly enclose and form a closed space for accommodating the battery cell, the second box comprises a second end wall, the second end wall is disposed opposite to the first end wall along a first direction, the second box has a first side surface in a second direction, and the first direction intersects with the second direction,

wherein one end of the first side wall is connected to the first end wall, and the other end of the first side wall is connected to the first side surface.

2. The battery according to claim **1**, wherein an inner surface of the first side wall is connected to the first side surface.

3. The battery according to claim **1**, wherein the second end wall has a first surface facing the first end wall, and one end that is of the first side wall and that is away from the first end wall exceeds the first surface along a direction of the first end wall pointing towards the second end wall.

4. The battery according to claim **1**, wherein the second end wall has a second surface facing away from the first end wall, and the one end that is of the first side wall and that is away from the first end wall does not exceed the second surface along the direction of the first end wall pointing towards the second end wall.

5. The battery according to claim **1**, wherein a material density of the first box is lower than a material density of the second box.

6. The battery according to claim **5**, wherein a material of the first box is plastic, and a material of the second box is aluminum alloy.

7. The battery according to claim **1**, wherein the battery further comprises a first fastener, and the first side wall is connected to the first side surface through the first fastener.

8. The battery according to claim **7**, wherein the first side wall is provided with a first through hole, the first side surface is provided with a first threaded hole, and the first fastener passes through the first through hole and is connected to the first threaded hole.

9. The battery according to claim **1**, wherein the battery further comprises a first sealing element, and the first sealing element is disposed between the first side wall and the first side surface.

10. The battery according to claim **1**, wherein a quantity of the first side walls is two, and the two first side walls are disposed opposite to each other along the second direction; the second end wall is located between the two first side walls, and the second end wall has the two first side surfaces disposed opposite to each other along the second direction; and the first side surface corresponds one-to-one to the first side wall.

11. The battery according to claim **1**, wherein the second box further comprises two second side walls, the two second side walls are disposed opposite to each other along the second direction and connected to the second end wall, and the battery cell is disposed between the two second side walls.

12. The battery according to claim **1**, wherein openings are respectively formed at two ends of the first box along a third direction; the second box further comprises two third side walls, the two third side walls are disposed opposite to each other along the third direction and connected to the second end wall, and the two third side walls respectively close the two openings; and the first direction, the second direction and the third direction intersect in pairs.

13. The battery according to claim **12**, wherein the third side wall comprises a third surface facing the closed space, a fourth surface facing away from the closed space and a second side surface connecting the third surface and the fourth surface; and the first box comprises two connecting portions, the two connecting portions are respectively located at the two ends of the first box along the third direction, the connecting portion peripherally forms the opening, and the connecting portion is connected to the second side surface.

14. The battery according to claim **13**, wherein the second side surface comprises a first flat surface, a second flat surface and a transition surface, the first flat surface is disposed at one end that is of the third side wall and that is facing away from the second end wall, the two second flat surfaces are respectively disposed at two ends of the third side wall along the second direction, and the transition surface connects the first flat surface and the second flat surface, so that the first flat surface smoothly transitions to the second flat surface.

15. The battery according to claim **14**, wherein the transition surface is a circular arc transition surface; and

the transition surface has a first edge in the third direction, the first edge is circular arc-shaped, and a radius of the first edge is greater than or equal to 10 mm.

16. The battery according to claim **14**, wherein the transition surface comprises a first transition surface and a second transition surface connected to each other, an edge of the first transition surface in the third direction is flat, and an edge of the second transition surface in the third direction is circular arc-shaped;

a quantity of the first transition surfaces is n, with n≥1, and a quantity of the second transition surfaces is n+1; and any one of the first transition surfaces is disposed between two adjacent second transition surfaces; and

a length of the first transition surface is greater than or equal to 6 mm.

17. The battery according to claim **13**, wherein the second side surface comprises a circular arc surface and a second flat surface, the two second flat surfaces are respectively disposed at two ends of the third side wall along the second direction, and one end of one of the two second flat surfaces and one end of the other of the two second flat surfaces face away from the second end wall and are connected through the circular arc surface;

the second flat surface is flush with the first side surface;

the second end wall comprises a first portion and a second portion, and the second portion is located on one side of the first portion and connected to the first portion along the second direction; and a size of the first portion in the second direction is greater than a size of the second portion in the second direction;

the second portion has a third side surface in the second direction, and the third side surface is flush with the second flat surface;

the first portion has a fourth side surface in the second direction, and the third side surface protrudes from the fourth side surface along the third direction; and

the first portion has a third flat surface in the third direction, and one end of the third flat surface is in circular arc transition connection with the third side surface, and the other end of the third flat surface is in circular arc transition connection with the fourth side surface along the second direction.

18. The battery according to claim **13**, wherein the second side surface comprises a first flat surface and a transition surface, the first flat surface is disposed at one end that is of the third side wall and that is facing away from the second end wall, and an end portion of the first flat surface smoothly transitions to a side surface of the second end wall through the transition surface;

the first box further comprises a fourth side wall, and the fourth side wall is adjacent to the first side wall; and one end of the fourth side wall is connected to the first end wall, the connecting portion is disposed at the other end of the fourth side wall, and the connecting portion protrudes from the fourth side wall in a direction facing away from the closed space;

the battery further comprises a second fastener, and the connecting portion is connected to the second side surface through the second fastener;

the connecting portion is provided with a second through hole, the second side surface is provided with a second threaded hole, and the second fastener passes through the second through hole and is connected to the second threaded hole;

the battery further comprises a second sealing element, and the second sealing element is disposed between the connecting portion and the second side surface;

the battery further comprises a first sealing element, and the first sealing element is disposed between the first side wall and the first side surface, wherein two ends of the first sealing element are respectively connected to the two second sealing elements; and

a size of one of the third side walls along the first direction is smaller than a size of the other one of the third side walls along the first direction.

19. The battery according to claim **1**, wherein the first box has a fifth side surface in the third direction, the second box has a sixth side surface in the third direction, the sixth side surface is disposed adjacent to the first side surface, the fifth side surface and the sixth side surface are connected to each other, and the first direction, the second direction and the third direction intersect in pairs;

the sixth side surface is in a circular arc transition to the first side surface; and

the second end wall is provided with a mounting portion for mounting the battery to an electrical device.

20. An electrical device, wherein the electrical device comprises the battery according to claim **1**, and the battery is configured to provide electrical energy.

* * * * *